

Can you recognise the music playing?

6COSC020W: APPLIED AI

WEEK 8: ARTIFICIAL NEURAL NETWORKS (ANNs)

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In the meantime, please respond to:
Pollev.com/esterbonmati

NOTES

- We will use PollEverywhere multiple times during the lecture: **polleverywhere.com/esterbonmati**
- It may be useful to have a notebook and pen/pencil (or your preferred method) to take some notes.
- Please engage during the lecture.

LEARNING OUTCOMES OF THIS SESSION

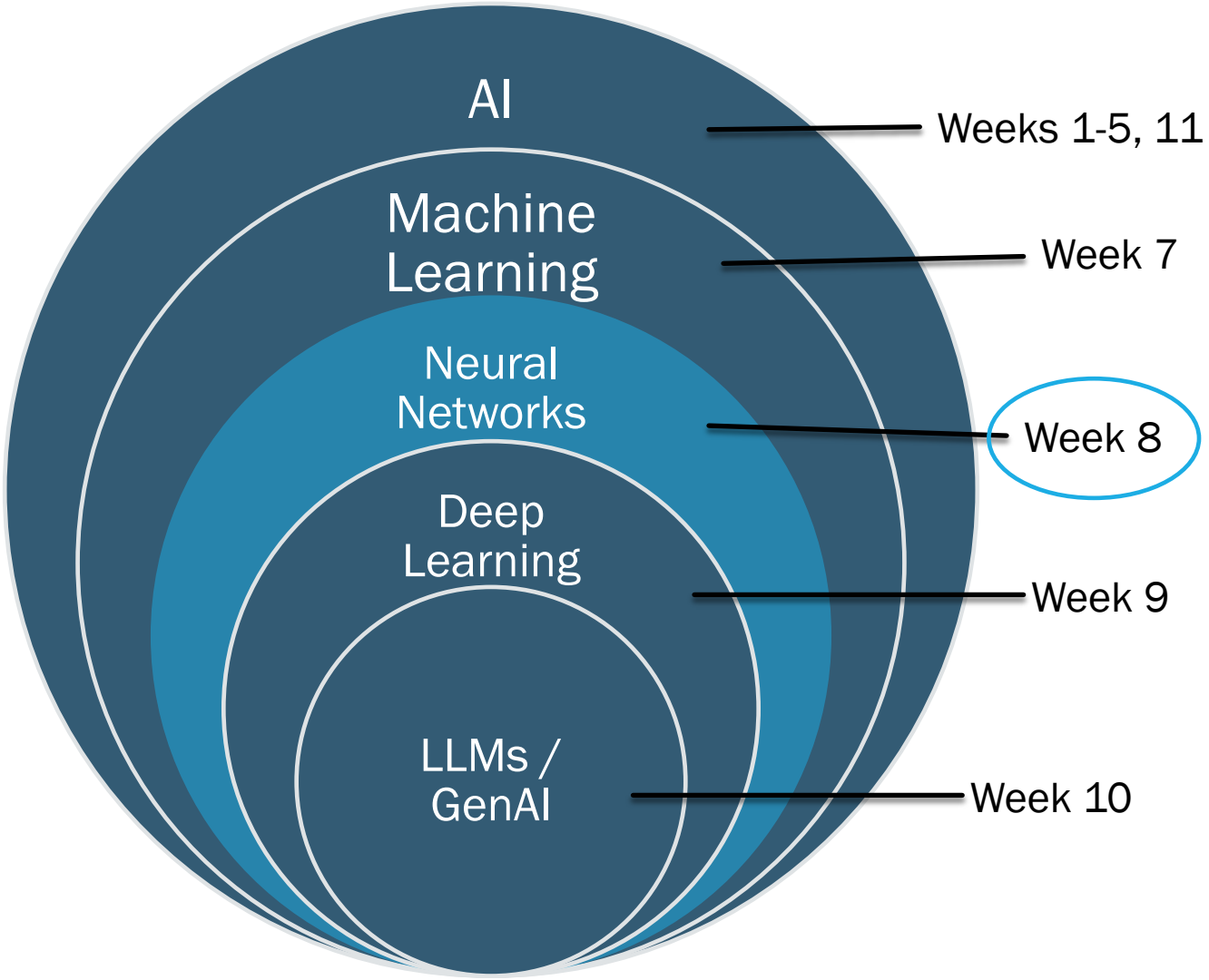
Perceptron

- Perceptron: Inputs, outputs and weights
- Forward propagation
- Explain what is an activation functions
- Be able to calculate the output of a perceptron

Neural Networks

- Be able to list the name of the different layers
- Be able to describe how neural networks work in your own words
- Explain problems with data
- Explain what are hyperparameters
- Be able to evaluate performance and calculate accuracy

ARTIFICIAL INTELLIGENCE: NEURAL NETWORKS



Before: what is an artificial neural network?

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Nobody has responded yet.

Hang tight! Responses are coming in.

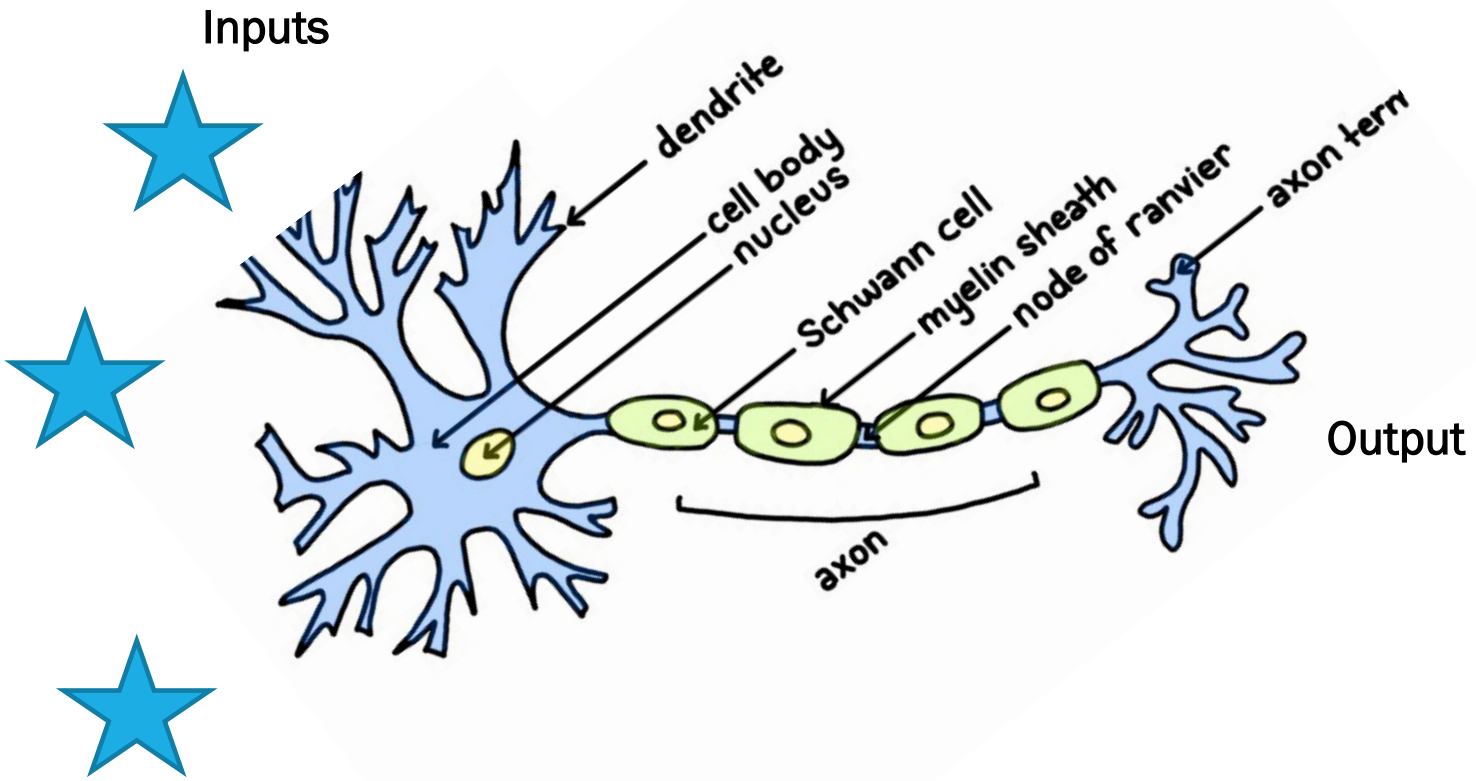
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

ARTIFICIAL NEURAL NETWORKS (ANNs)

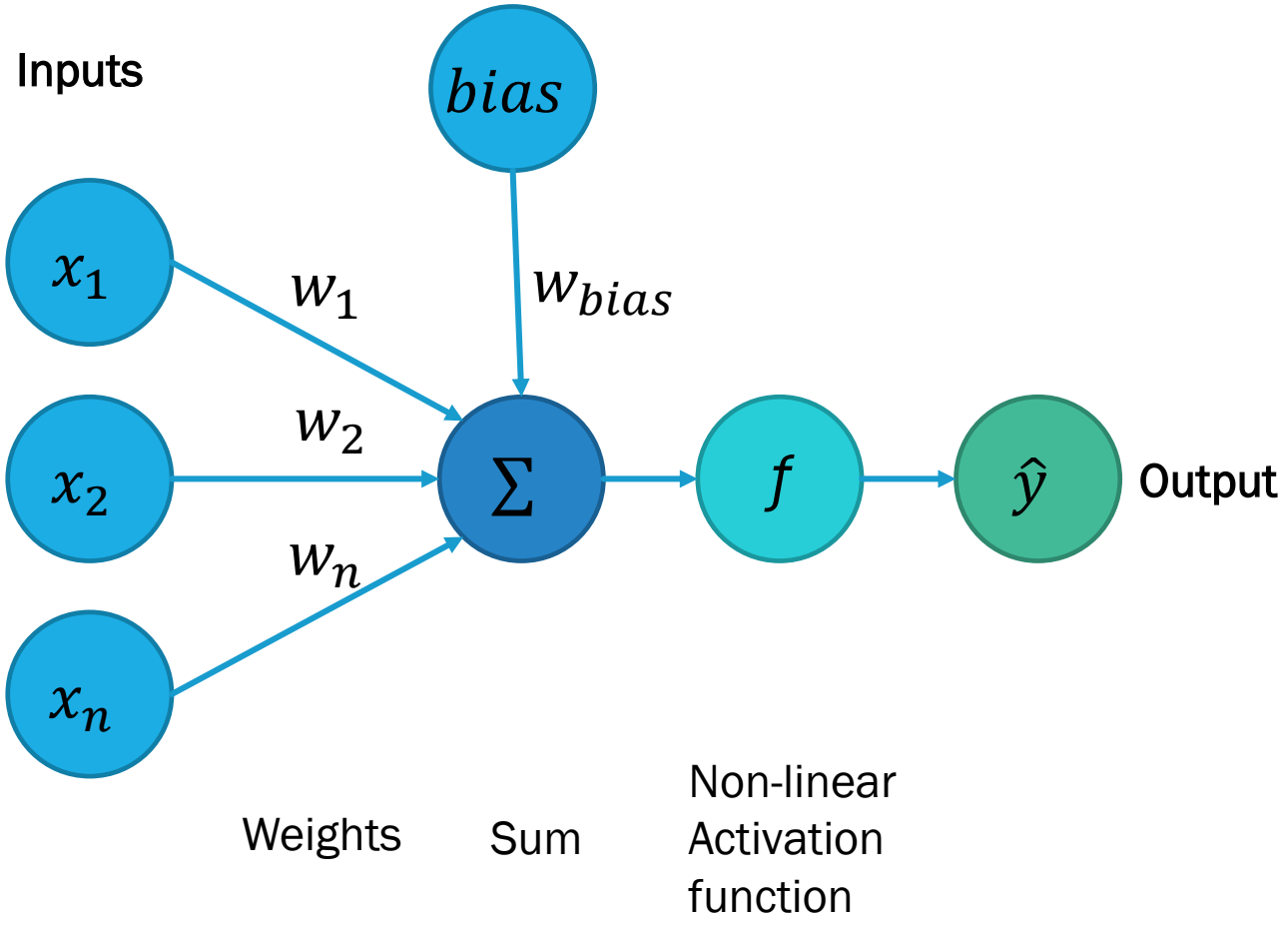
Why are they important?
ANN can **help computers make intelligent decisions with limited human assistance.**
They can learn and model the **relationships** between input and output data that are nonlinear and complex.



NEURON

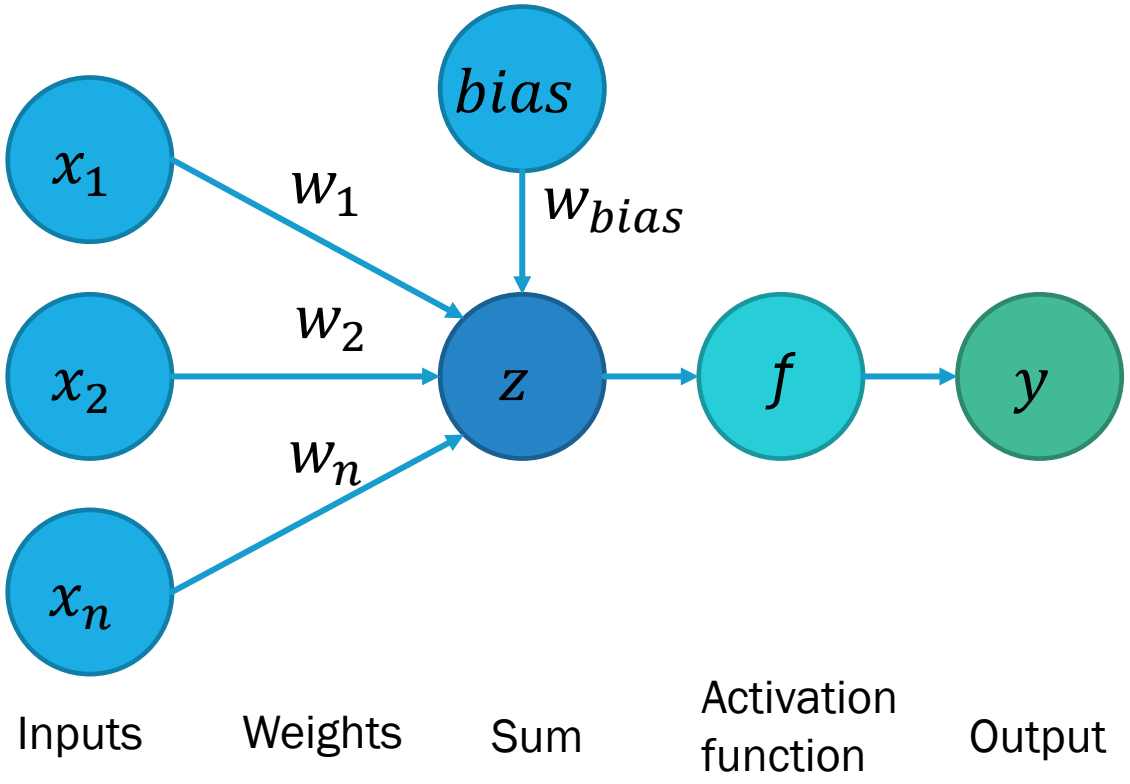

Inputs
dendrite
cell body
nucleus
Schwann cell
myelin sheath
node of ranvier
axon term
Output
axon

THE PERCEPTRON



THE PERCEPTRON

- Building block of neural networks (single neuron)
- Mimics the neuron in the human brain
- **Forward propagation (input → neuron → output)**



Output
Bias
Inputs

$$y = f\left(bias\ w_{bias} + \sum_{i=1}^{n} x_i\ w_i\right)$$

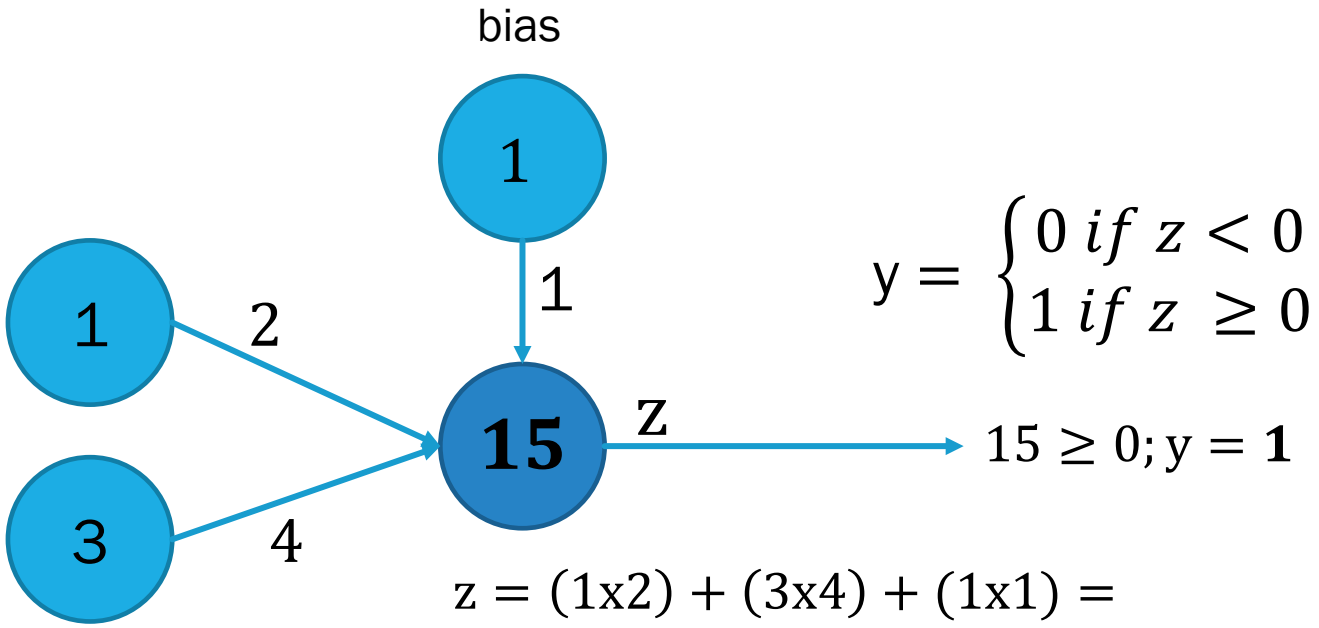
Non-linear function
Sum
Weights

$$z = x_1\ w_1 + x_2\ w_2 + \ldots + x_n\ w_n + bias\ w_{bias}$$

$$y = f(z)$$

Example: $y = \begin{cases} 0\ if\ z < 0 \\ 1\ if\ z \geq 0 \end{cases}$

PERCEPTRON: FORWARD PROPAGATION EXAMPLE



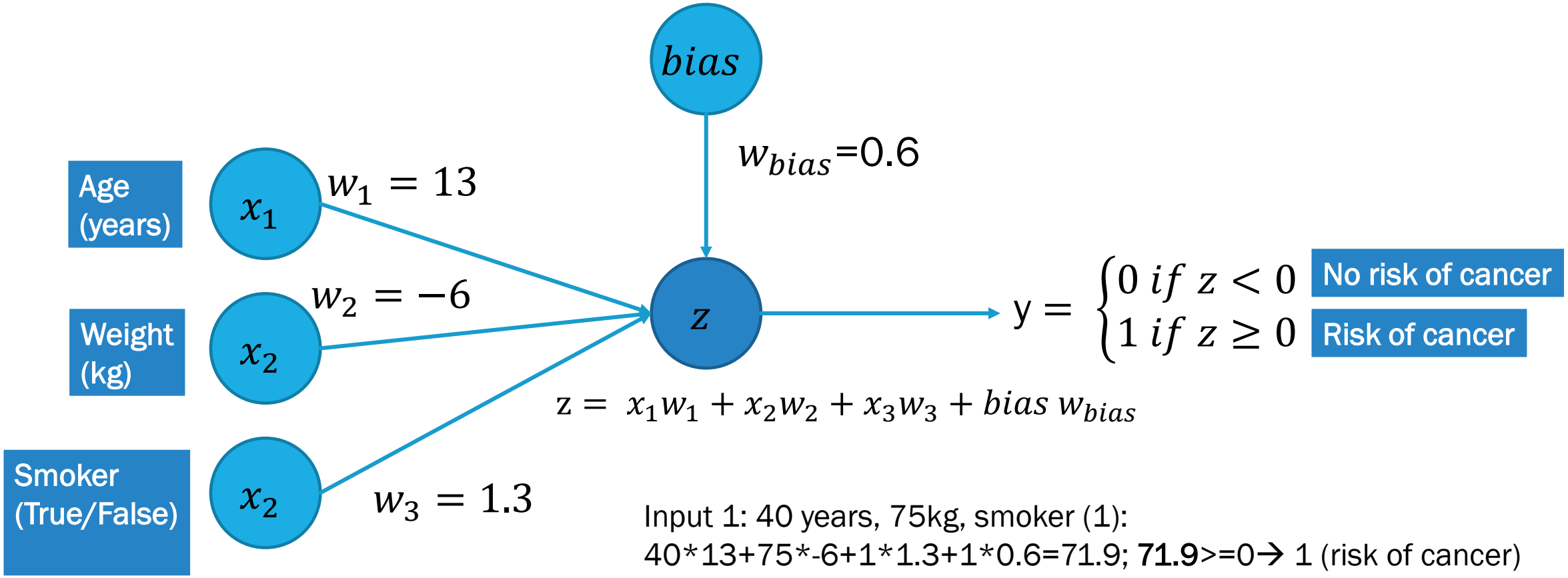
$$y = \begin{cases} 0 \; if \; z < 0 \\ 1 \; if \; z \geq 0 \end{cases}$$

$15 \geq 0; y = \mathbf{1}$

$z = (1x2) + (3x4) + (1x1) =$

$= 2 + 12 + 1 = \mathbf{15}$

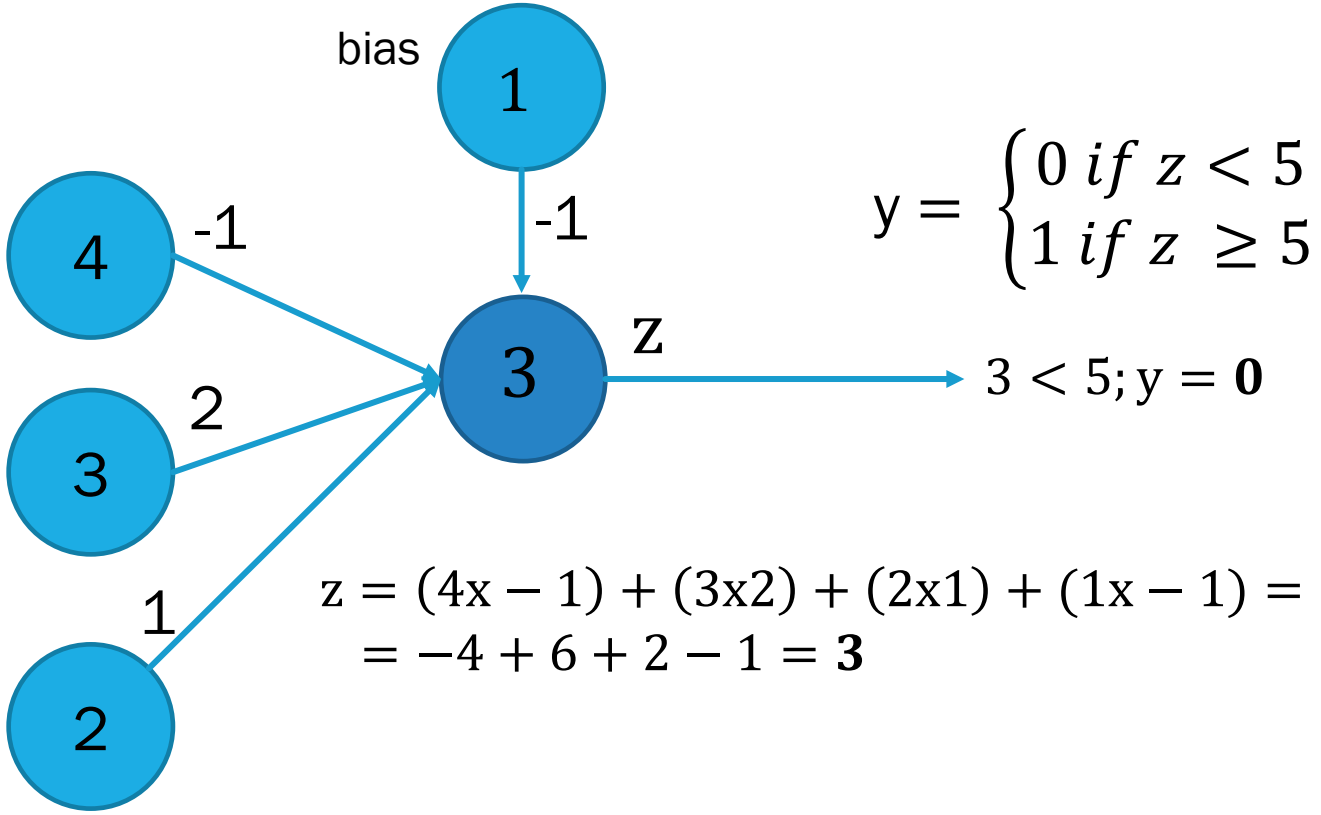
PERCEPTRON: FORWARD PROPAGATION EXAMPLE



Input 1: 40 years, 75kg, smoker (1):
40*13+75*-6+1*1.3+1*0.6=71.9; **71.9**>=0→ 1 (risk of cancer)

Input 2: 30 years, 70kg, non-smoker (0):
30*13+70*-6+0*1.3+1*0.6=-29.4; **-29.4**<0→ 0 (no risk of cancer)

PERCEPTRON: TEST 1



$$y = \begin{cases} 0 \; if \; z < 5 \\ 1 \; if \; z \geq 5 \end{cases}$$

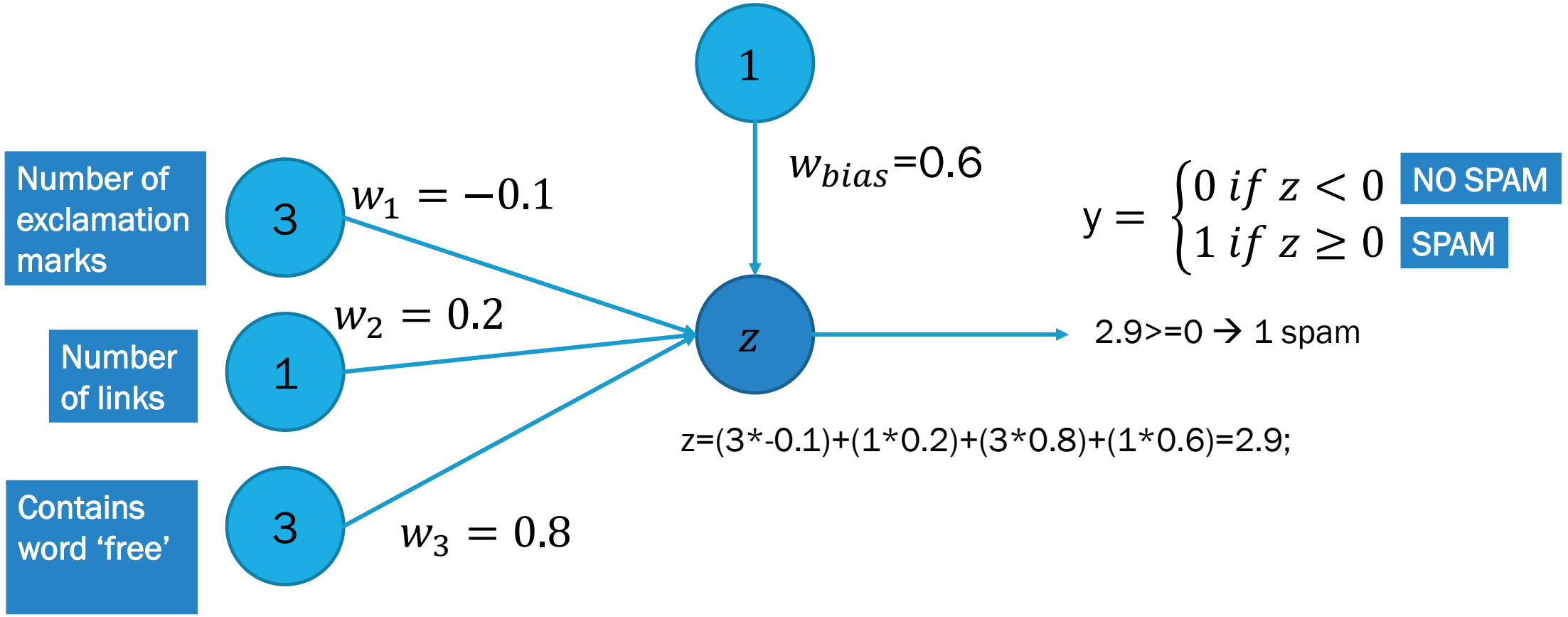
$3 < 5; y = \mathbf{0}$

$$z = (4x - 1) + (3x2) + (2x1) + (1x - 1) =$$
$$= -4 + 6 + 2 - 1 = \mathbf{3}$$

Submit your answer here: pollev.com/esterbonmati

Compare answers with the person sitting next to you

PERCEPTRON: TEST 2 SPAM DETECTOR

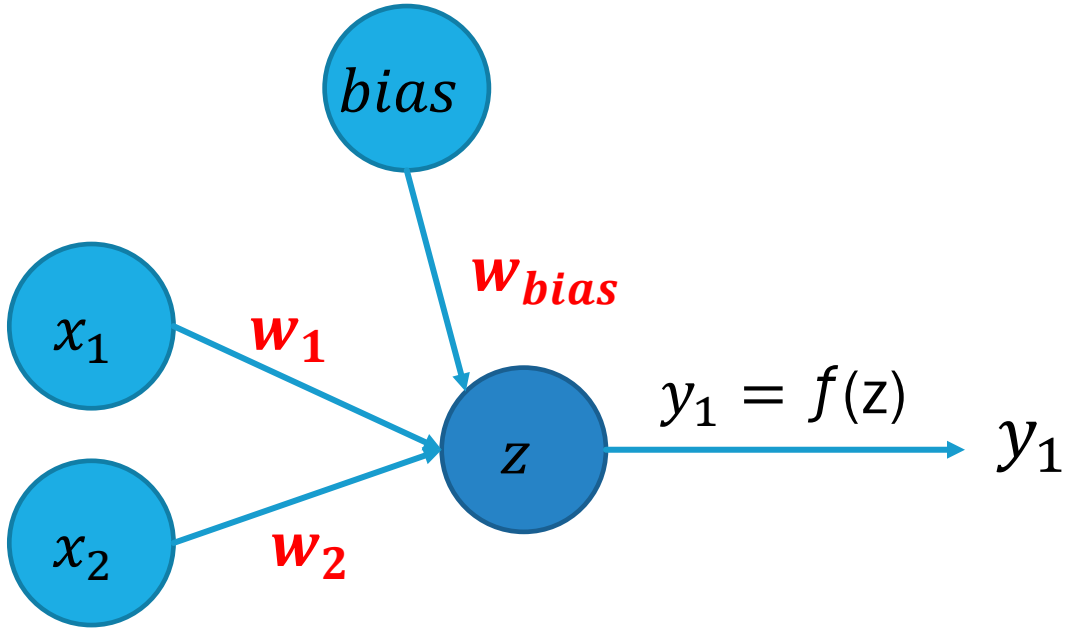


Submit your answer here: pollev.com/esterbonmati

Compare answers with the person sitting next to you

PERCEPTRON: HOW DO WE FIND THE WEIGHTS?

Problem: we want to train a perceptron to classify points in 2 classes.

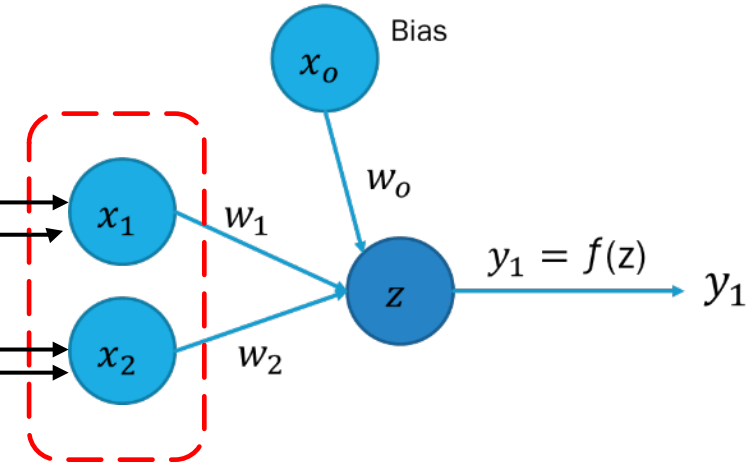


How do we find the weights to classify the points?

$$z = x_1 w_1 + x_2 w_2 + bias w_{bias}$$

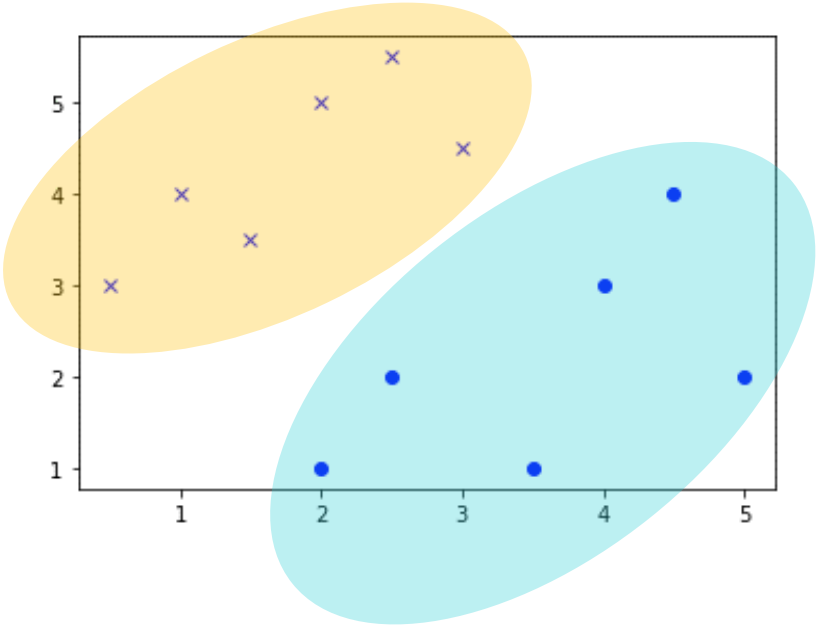
PERCEPTRON: INPUT DEFINITION

Problem: We have 2 groups of samples, each of them defined by 2 values, and we want to use a perceptron to classify them into their group.



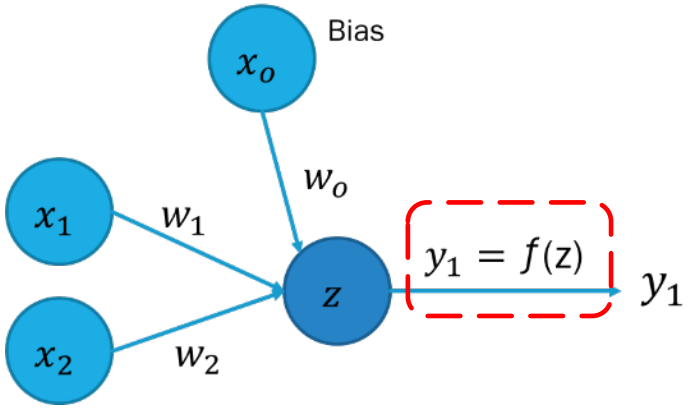

```
x1_group1 = [2, 5, 4, 2.5, 3.5, 4.5]
x2_group1 = [1, 2, 3, 2, 1, 4]

x1_group0 = [0.5, 2.5, 3, 1.5, 1, 2]
x2_group0 = [3, 5.5, 4.5, 3.5, 4, 5]
```



PERCEPTRON: ACTIVATION FUNCTION

$$y = \begin{cases} -1 \; if \; z \leq 0 \\ 1 \; if \; z > 0 \end{cases}$$



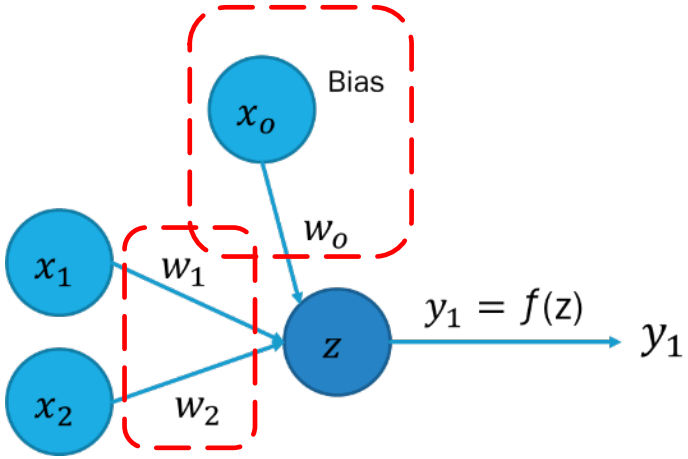
If the sum is is greater than 0 we return 1, otherwise we return 0

In [112]:
```
def activation_function(value):
    if value > 0: return 1
    else: return -1
```

PERCEPTRON: WEIGHTS AND BIAS DEFINITION

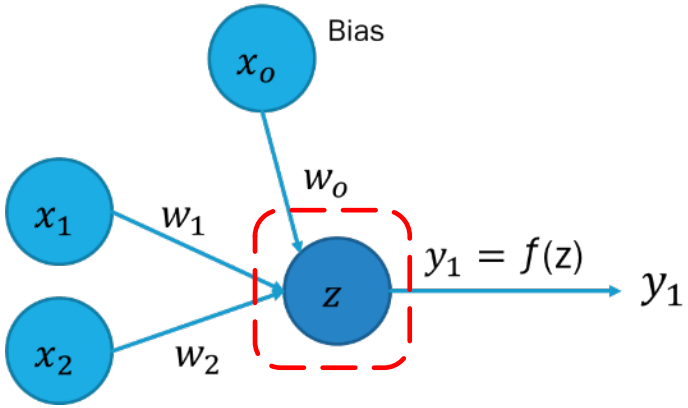
Some random weights…

```
w1 = 0.75
w2 = -0.5
bias = 1
wbias = 1
```



How good are these weights in grouping the input data?

PERCEPTRON: FORWARD PROPAGATION



We need to find z in order to calculate the output y …

```
def perceptron(x1, x2, bias):
    z = x1*w1 + x2*w2 + bias*wbias
    return z
```

Then we can calculate the output y using the perceptron and the activation function:

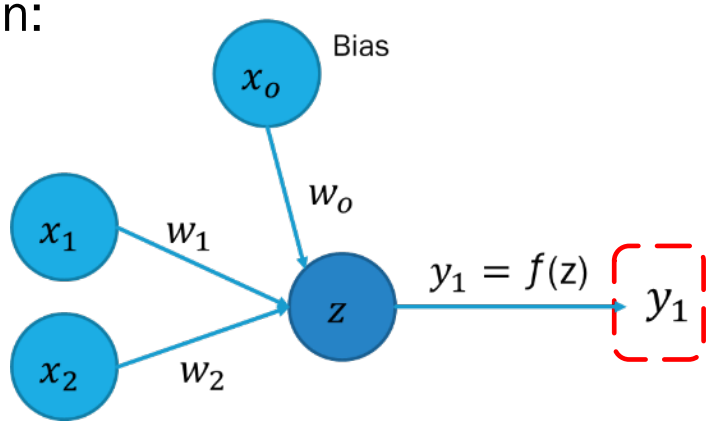
```
for i in range(len(x1_group0)):
    z = perceptron(x1_group0[i], x2_group0[i], bias)
    group = activation_function(z)
    print('Point ' + str(i) + ' in group 0 returned ' + str(group))
```

PERCEPTRON: PREDICTION AND RESULTS

We calculate the output for each group, so we know which is the prediction:

```
    group = activation_function(z)
    print('Point ' + str(i) + ' in group 1 returned ' + str(group))

for i in range(len(x1_group0)):
    z = perceptron(x1_group0[i], x2_group0[i], bias)
    group = activation_function(z)
    print('Point ' + str(i) + ' in group 0 returned ' + str(group))
```




```
Point 0 in group 1 returned 1
Point 1 in group 1 returned 1
Point 2 in group 1 returned 1
Point 3 in group 1 returned 1
Point 4 in group 1 returned 1
Point 5 in group 1 returned 1
Point 0 in group 0 returned -1
Point 1 in group 0 returned 1
Point 2 in group 0 returned 1
Point 3 in group 0 returned 1
Point 4 in group 0 returned -1
Point 5 in group 0 returned -1
```

Incorrect

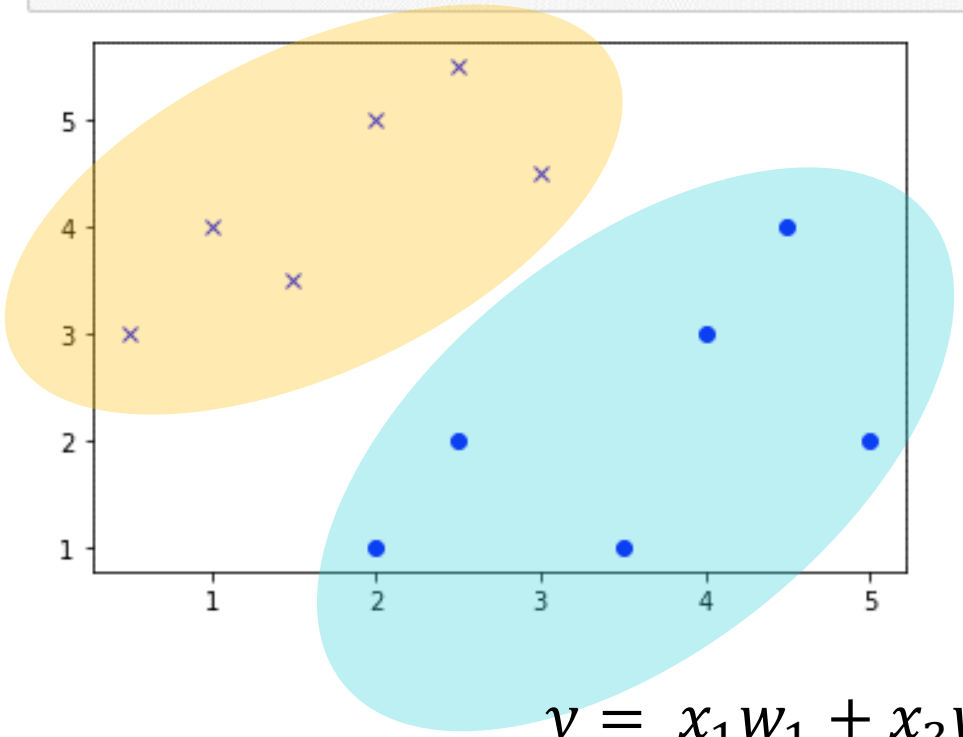
$$accuracy = \frac{number\ correct\ samples}{total\ number\ of\ samples} = \frac{9}{12} = 0.75 \rightarrow \mathbf{75}\ \%$$

Can we improve the accuracy if we change the weights?

PERCEPTRON: VISUALISATION

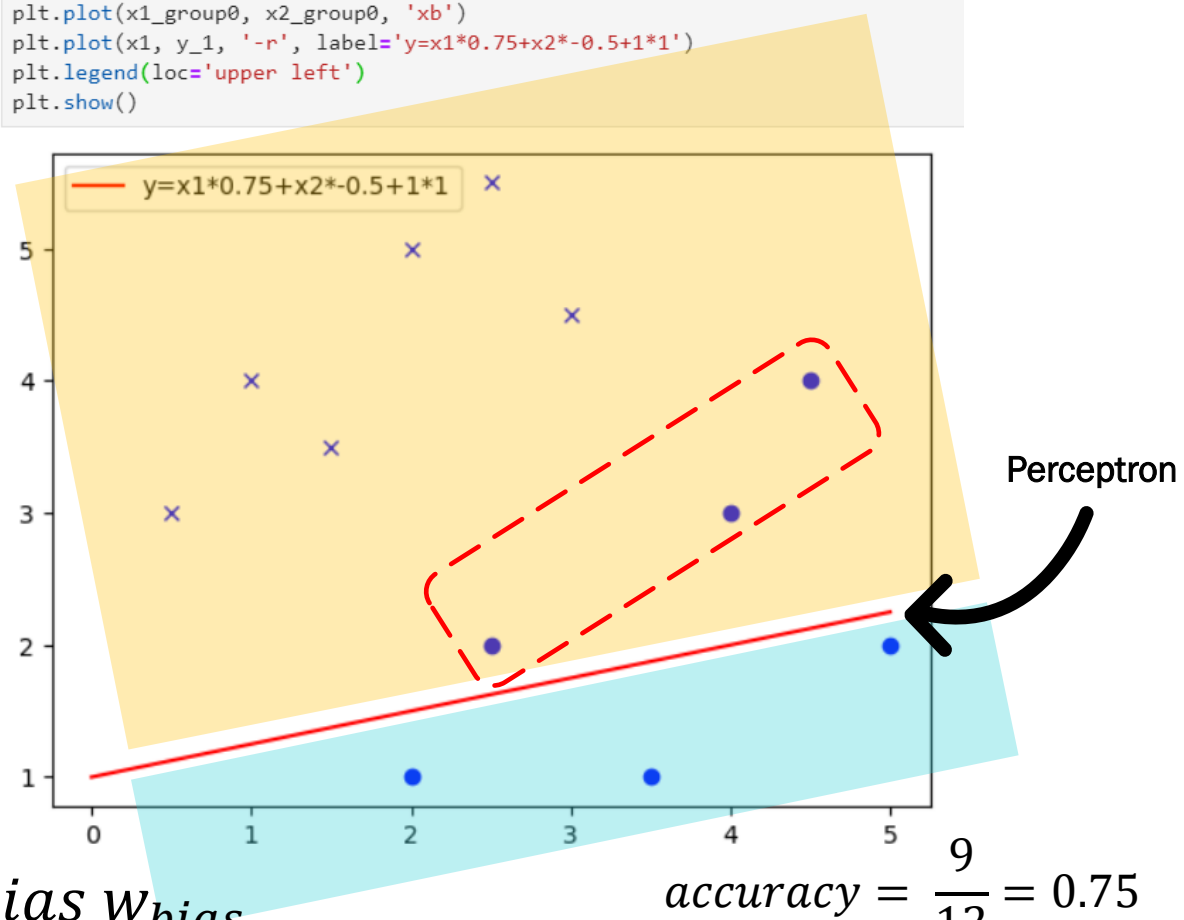
Plot the points

In [117]:
```
fig = plt.figure()
ax = plt.axes()
plt.plot(x1_group1, x2_group1, 'ob')
plt.plot(x1_group0, x2_group0, 'xb')
plt.show()
```



[7]:
```
x1 = np.linspace(0,5,100)
x2 = np.linspace(0,5,100)
y_1 = w1*x1 + w2*x2 + 1

plt.plot(x1_group1, x2_group1, 'ob')
plt.plot(x1_group0, x2_group0, 'xb')
plt.plot(x1, y_1, '-r', label='y=x1*0.75+x2*-0.5+1*1')
plt.legend(loc='upper left')
plt.show()
```



$$y = x_1 w_1 + x_2 w_2 + bias\ w_{bias}$$

$$accuracy = \frac{9}{12} = 0.75$$

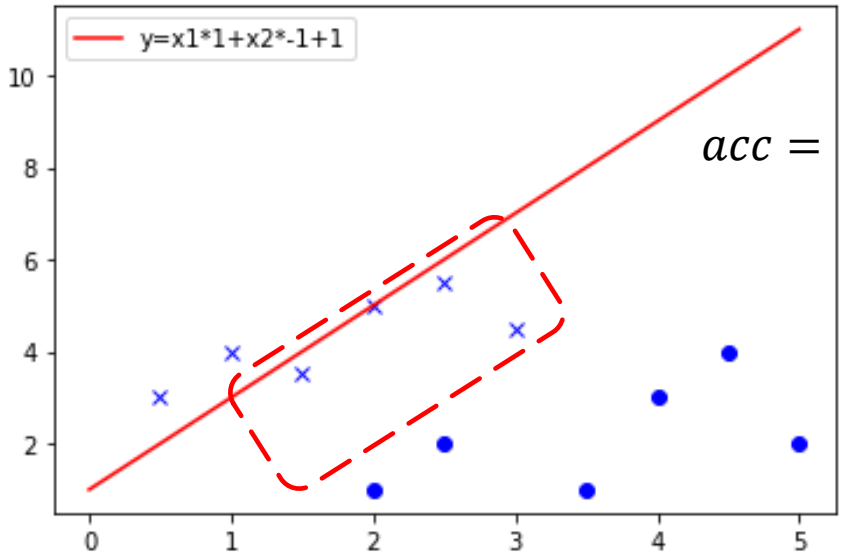
PERCEPTRON: UPDATE WEIGHTS

To change the prediction output, we need to change the weights

In [102]:
```
w1 = 1
w2 = 1
bias = 1
wbias = 1

y_1 = w1*x1 + w2*x2 + bias*wbias

plt.plot(x1_group1, x2_group1, 'ob')
plt.plot(x1_group0, x2_group0, 'xb')
plt.plot(x1, y_1, '-r', label='y=x1*1+x2*1+1*1')
plt.legend(loc='upper left')
plt.show()
```

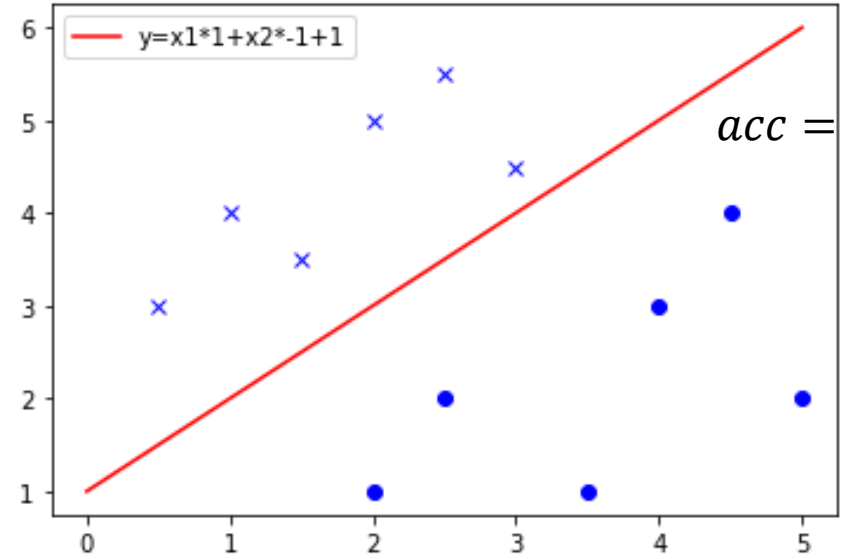


$$acc = \frac{8}{12} = 0.67$$

In [103]:
```
w1 = 3
w2 = -2
bias = 1
wbias = -1

y_1 = x1*w1 + x2*w2 + bias*bias

plt.plot(x1_group1, x2_group1, 'ob')
plt.plot(x1_group0, x2_group0, 'xb')
plt.plot(x1, y_1, '-r', label='y=x1*3+x2*-2+1*-1')
plt.legend(loc='upper left')
plt.show()
```



$$acc = \frac{12}{12} = 1.0$$

PERCEPTRON: NEW PREDICTION

In [104]:
```
for i in range(len(x1_group1)):
    sum = perceptron(x1_group1[i], x2_group1[i], bias)
    group = activation_function(sum)
    print('Point ' + str(i) + ' in group 1 returned ' + str(group))

for i in range(len(x1_group0)):
    sum = perceptron(x1_group0[i], x2_group0[i], bias)
    group = activation_function(sum)
    print('Point ' + str(i) + ' in group 0 returned ' + str(group))
```

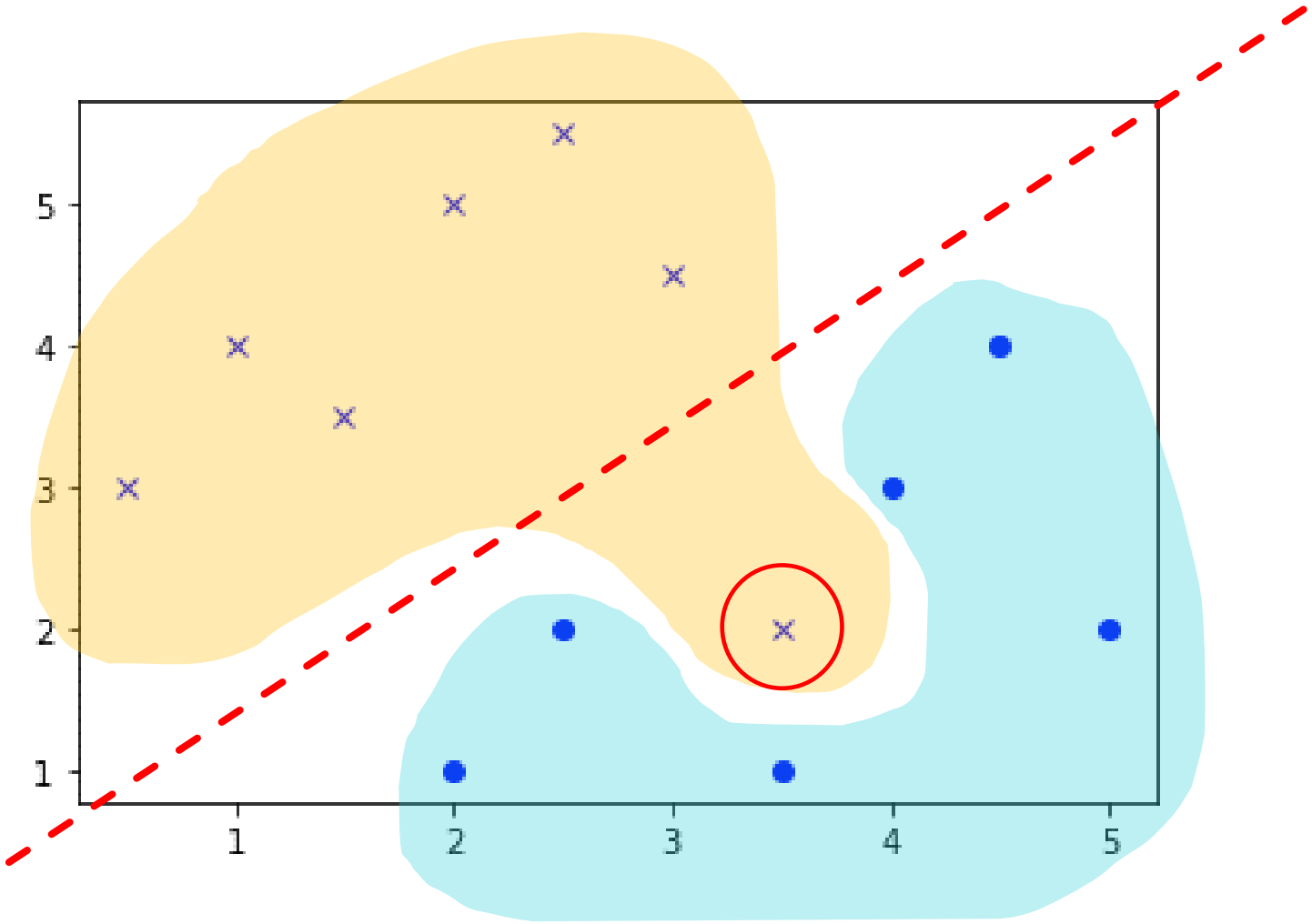
```
Point 0 in group 1 returned 1
Point 1 in group 1 returned 1
Point 2 in group 1 returned 1
Point 3 in group 1 returned 1
Point 4 in group 1 returned 1
Point 5 in group 1 returned 1
Point 0 in group 0 returned -1
Point 1 in group 0 returned -1
Point 2 in group 0 returned -1
Point 3 in group 0 returned -1
Point 4 in group 0 returned -1
Point 5 in group 0 returned -1
```





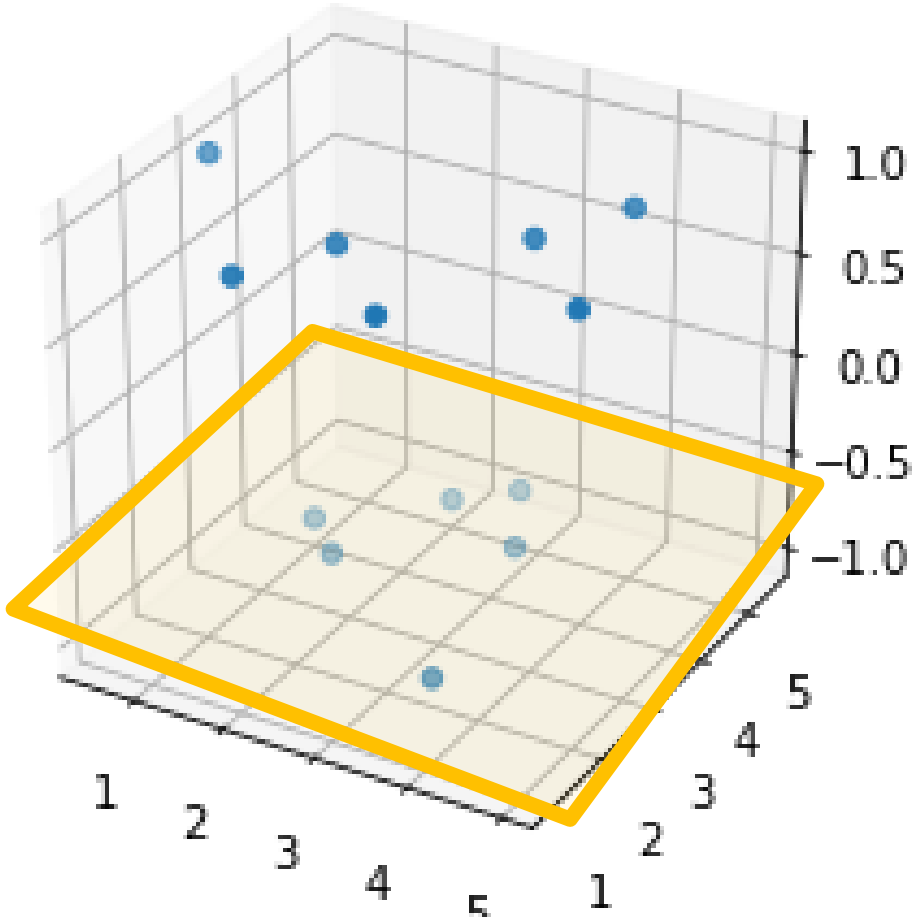
$$acc = \frac{12}{12} = 1.0 \rightarrow 100\ \%$$

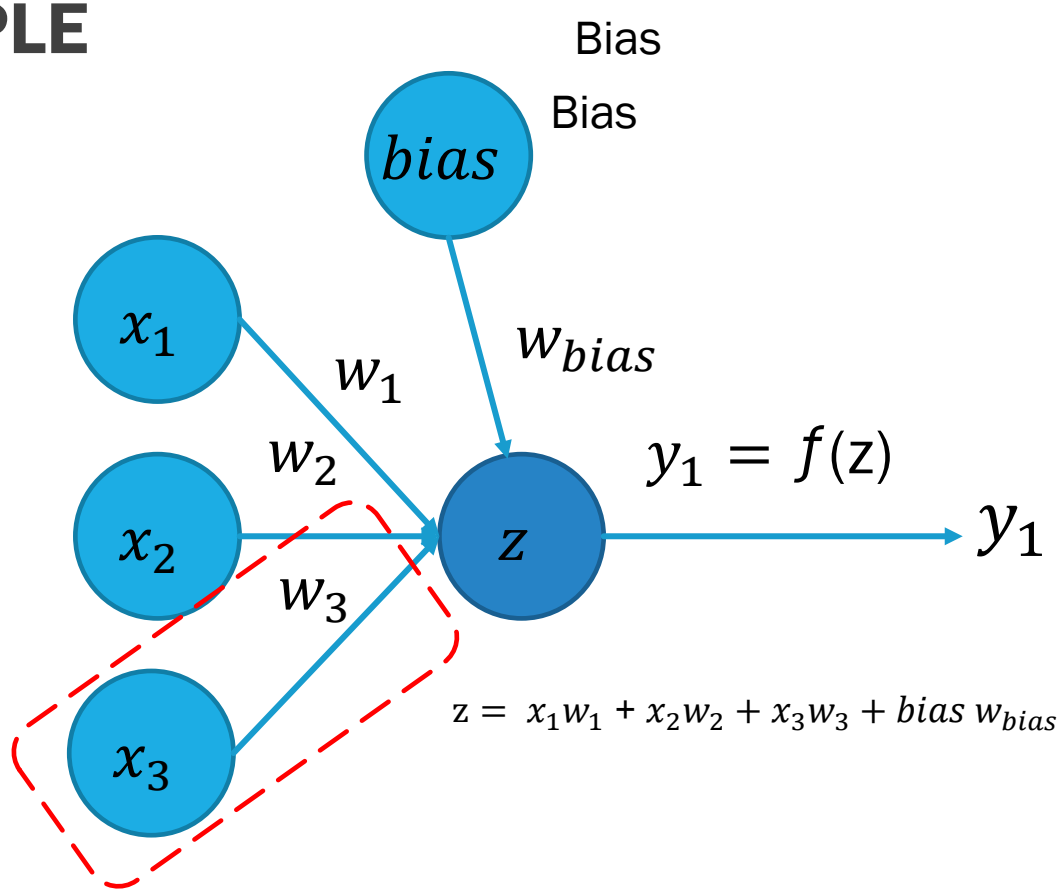
NEURAL NETWORKS: ANOTHER EXAMPLE



NEURAL NETWORKS: ANOTHER EXAMPLE

$$y = x_1 w_1 + x_2 w_2 + x_3 w_3 + b$$





[For reference only] Equation of a plane = $ax + by + cz + d$

Do you know another example of a perceptron?

- Logic operations (**except XOR**)
- Linear separation
- Predict who is going to pass this module based on the number of lectures and tutorials attended
- ?





When poll is active respond at **PollEv.com/esterbonmati**

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How many inputs can a perceptron have?

2

0%

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How many weights has each input in a perceptron?

0

0%

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Applied AI

Moderate | Visual settings | Edit

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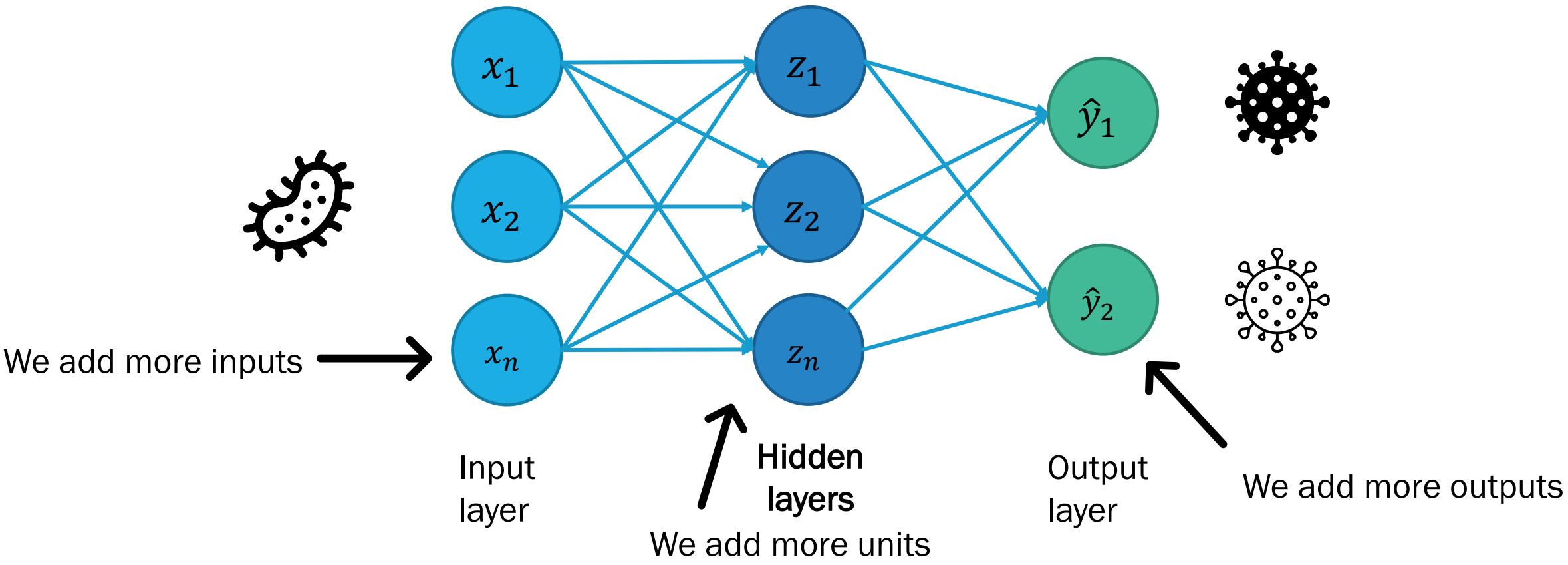
Q&A

Nobody has responded yet.

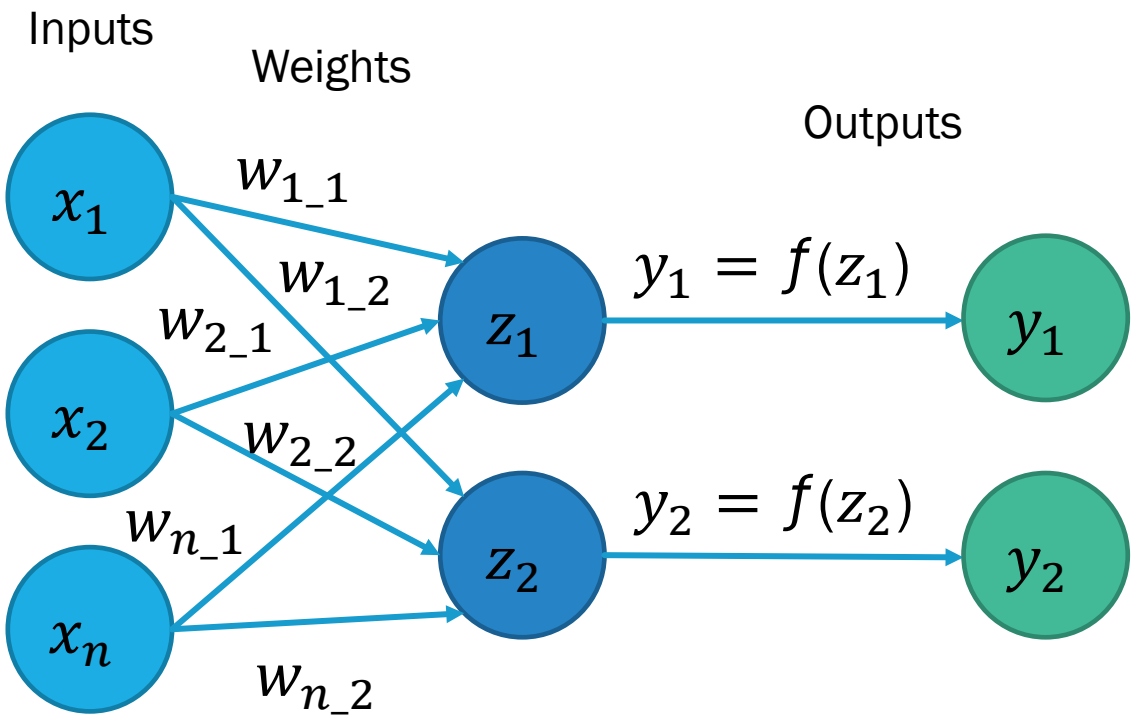
Hang tight! Responses are coming in.

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MULTI-LAYER NEURAL NETWORKS



ARTIFICIAL NEURAL NETWORKS (ANN)



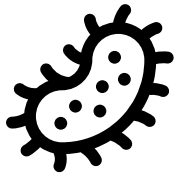
Each output has its own weight for each input

$$z = w_0 + \sum_{i=1}^{n} x_i \, w_i$$

$$\hat{y} = f\left(w_0 + \sum_{i=1}^{n} x_i \, w_i \right)$$

MULTI-LAYER NEURAL NETWORKS EXAMPLE

Input

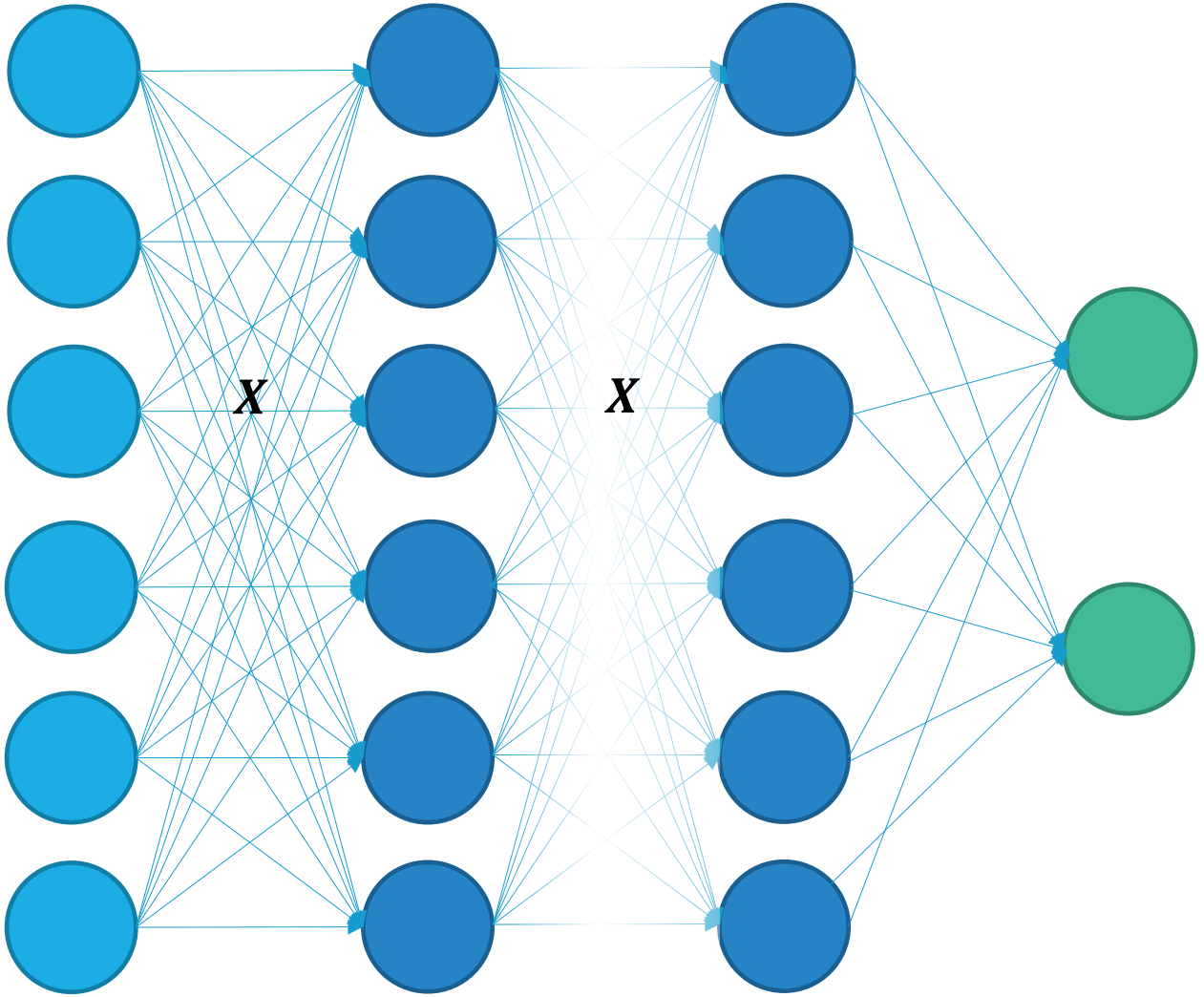


data

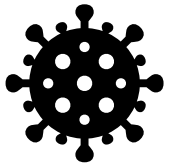
Input layer

Hidden layers

Output layer

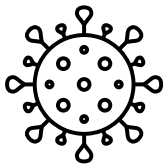


Dense layer
==
Fully connected layer



Probability of cancer 95%

Probability of no cancer 5%



ANNs: HOW MANY INPUTS AND OUTPUTS DO WE NEED?

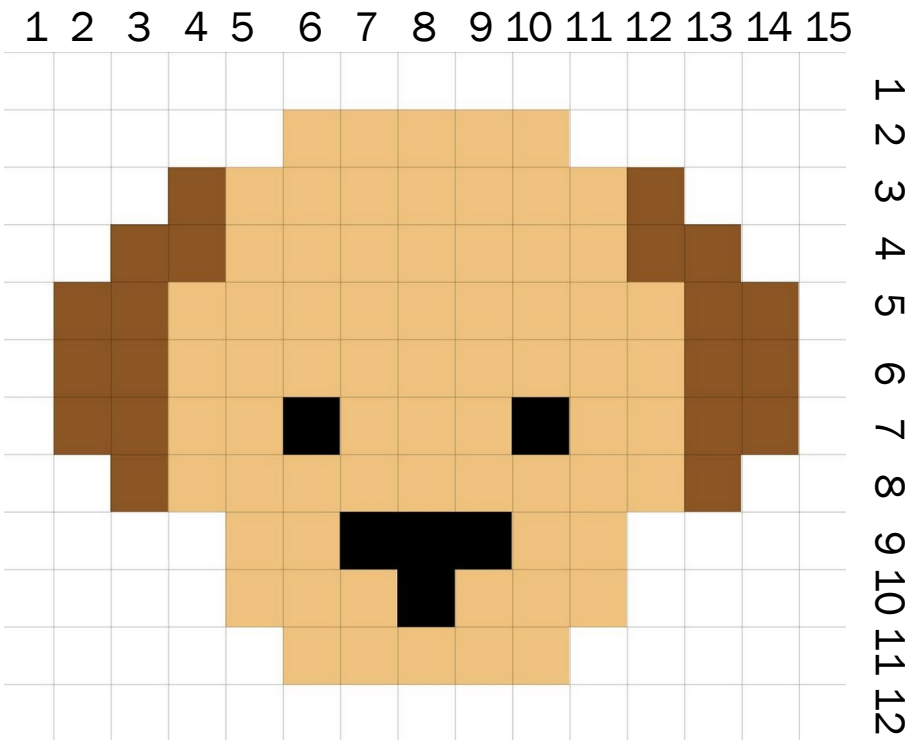
- The number of inputs will depend on our data.
- Example 1: Iris dataset

4 inputs | 1 output

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
50	7.0	3.2	4.7	1.4	versicolor
51	6.4	3.2	4.5	1.5	versicolor
52	6.9	3.1	4.9	1.5	versicolor
53	5.5	2.3	4.0	1.3	versicolor
54	6.5	2.8	4.6	1.5	versicolor
55	5.7	2.8	4.5	1.3	versicolor

ANNs: HOW MANY INPUTS AND OUTPUTS DO WE NEED?

- Example 2: Images



→ 15x12 = 180 pixels → 180 input units

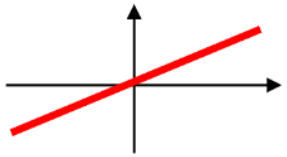
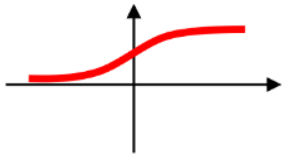
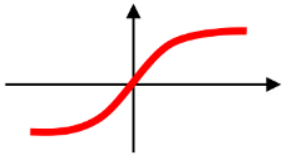
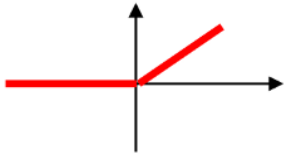
Dog/no dog → 1 label/class to predict → 1 output (0 or 1)

Next week: Convolutional Neural Networks

HOW MANY HIDDEN LAYERS DO WE NEED?

- The number of hidden layers, and the number of units depends on the complexity of the problem and there is not strict rule on determining the number of hidden layers.
- The more complex the problem, the more layers and units you will need.
- Easy problem: classify the iris dataset (4 input variables, 1 class as output)
- Complex problem: image recognition (225X255 images, 1000 classes as output)

FOR REFERENCE ONLY: ACTIVATION FUNCTIONS

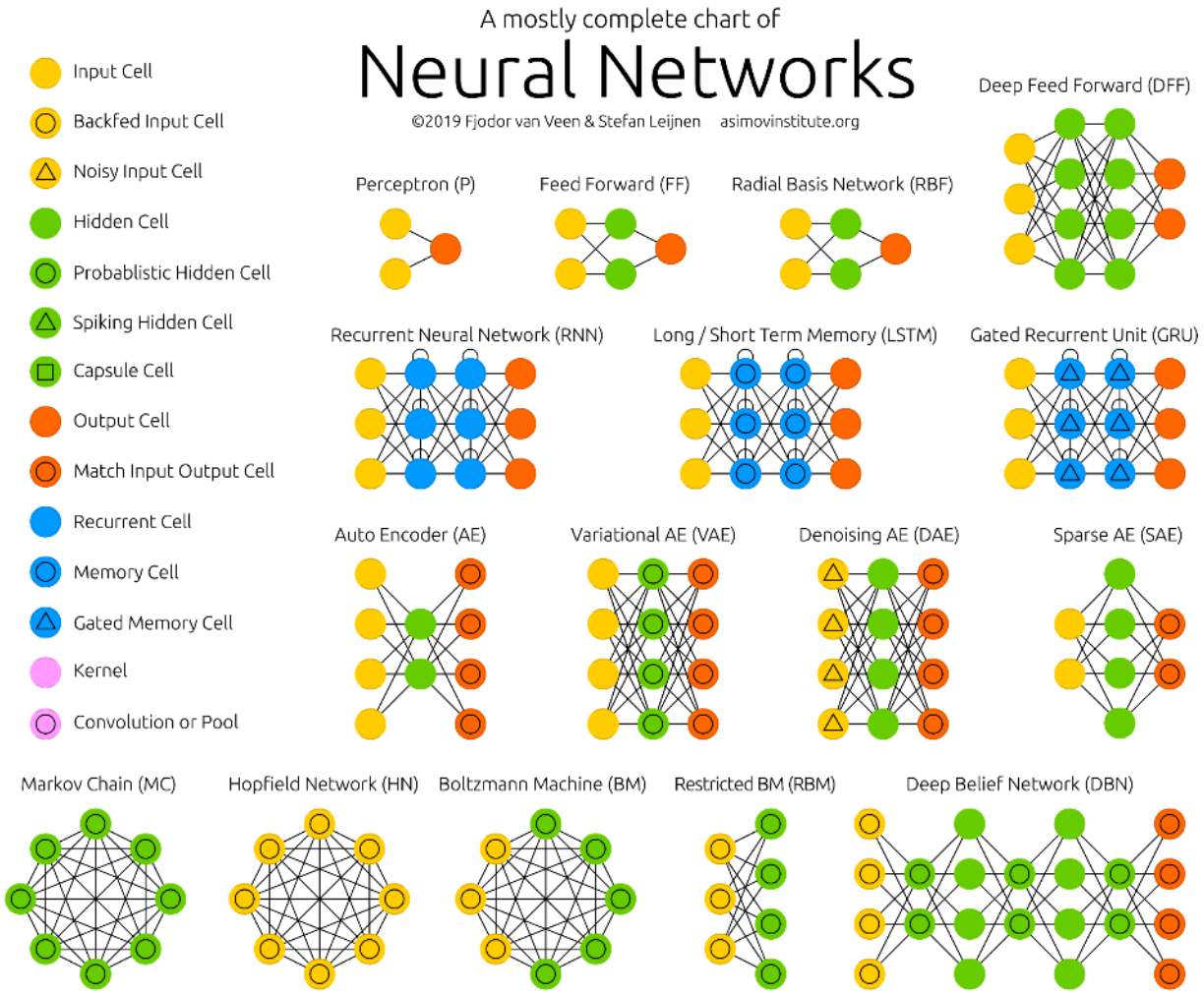
Activation function	Equation	Example	1D Graph
Linear	$\phi(z) = z$	Adaline, linear regression	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent (Tanh)	$\phi(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = max(0, z)$	Multi-layer Neural Networks	

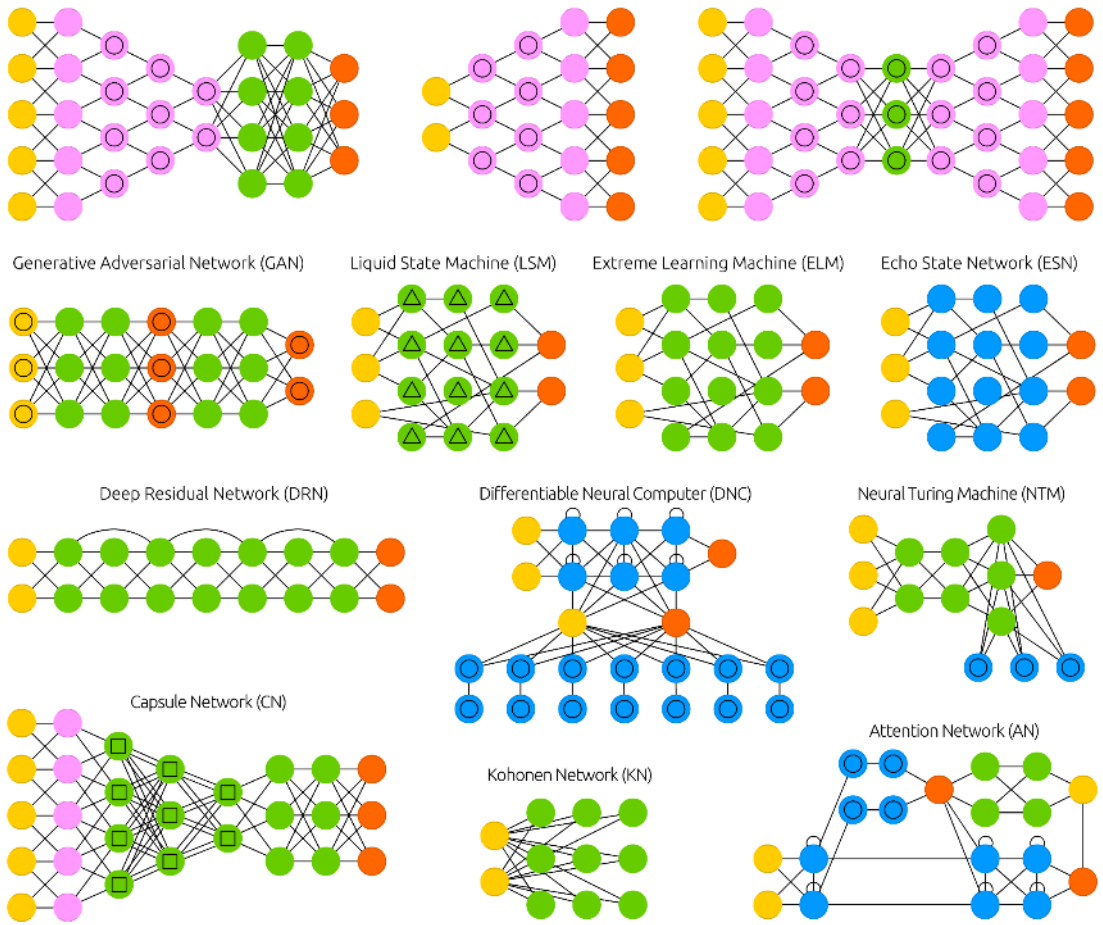
Copyright © Sebastian Raschka 2016
(http://sebastianraschka.com)

Non-linear

Non-linear activation functions allow to learn more complex mappings

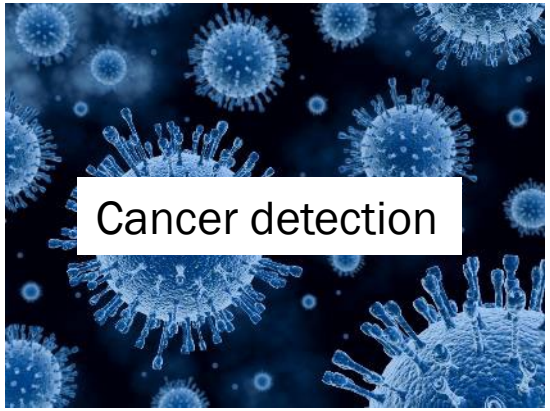
EXAMPLES OF NEURAL NETWORKS





Pictures from: https://www.asimovinstitute.org/neural-network-zoo/

WHAT CAN WE DO WITH ANN?



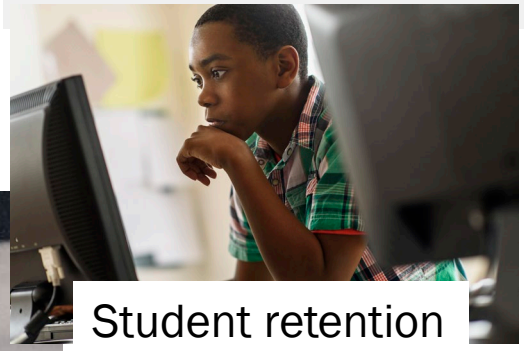
Shopping prediction

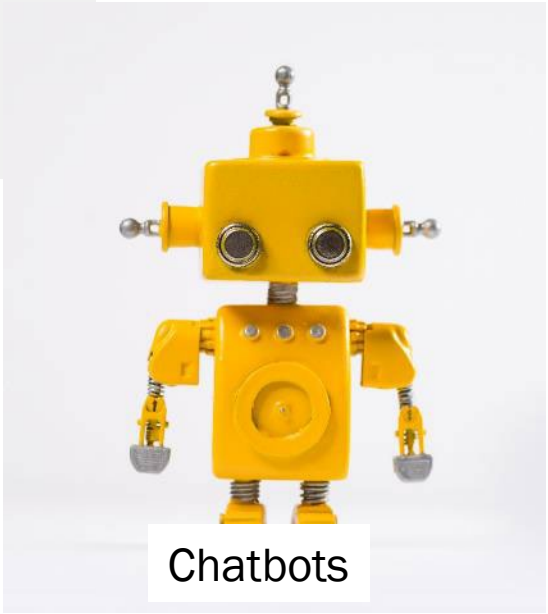




Recognition/Classification













Stock market prediction

Do you know more examples?

RESEARCH

Forecasting Climatic Trends Using Neural Networks: An Experimental Study Using Global Historical Data

Takeshi Ise [1,2] and Yurika Oba [1]*

[1] Field Science Education and Research Center (FSERC), Kyoto University, Kyoto, Japan, [2] Japan Science and Technology Agency (JST), Kawaguchi, Japan

Review Article

Artificial neural networks in the cancer genomics frontier

Andrew Oustimov[1], Vincent Vu[2]

[1]Department of Epidemiology & Biostatistics, College of Public Health, University of South Florida, Tampa, FL 33620, USA; [2]Department of Mathematics and Statistics, University of California, Los Angeles, CA, USA

Correspondence to: Andrew Oustimov, MPH. Department of Epidemiology & Biostatistics, College of Public Health, University of South Florida, 13201 Bruce B Downs Blvd, Tampa, FL 33620, USA. Email: aoustimo@mail.usf.edu.

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, VOL. 32, NO. 10, OCTOBER 2021 4291

Attention in Natural Language Processing

Andrea Galassi, Marco Lippi, and Paolo Torroni

AUTOMATIC LANGUAGE IDENTIFICATION USING DEEP NEURAL NETWORKS

Ignacio Lopez-Moreno[1], Javier Gonzalez-Dominguez[1,2], Oldrich Plchot[3], David Martinez[4], Joaquin Gonzalez-Rodriguez[2], Pedro Moreno[1]

[1]Google Inc., New York, USA
[2]ATVS-Biometric Recognition Group, Universidad Autonoma de Madrid, Spain
[3]Brno University of Technology, Czech Republic
[4]Aragon Institute for Engineering Research (I3A), University of Zaragoza, Spain
{elnota,jgd}@google.com

Tran *et al. Genome Medicine* (2021) 13:152
https://doi.org/10.1186/s13073-021-00968-x

Genome Medicine

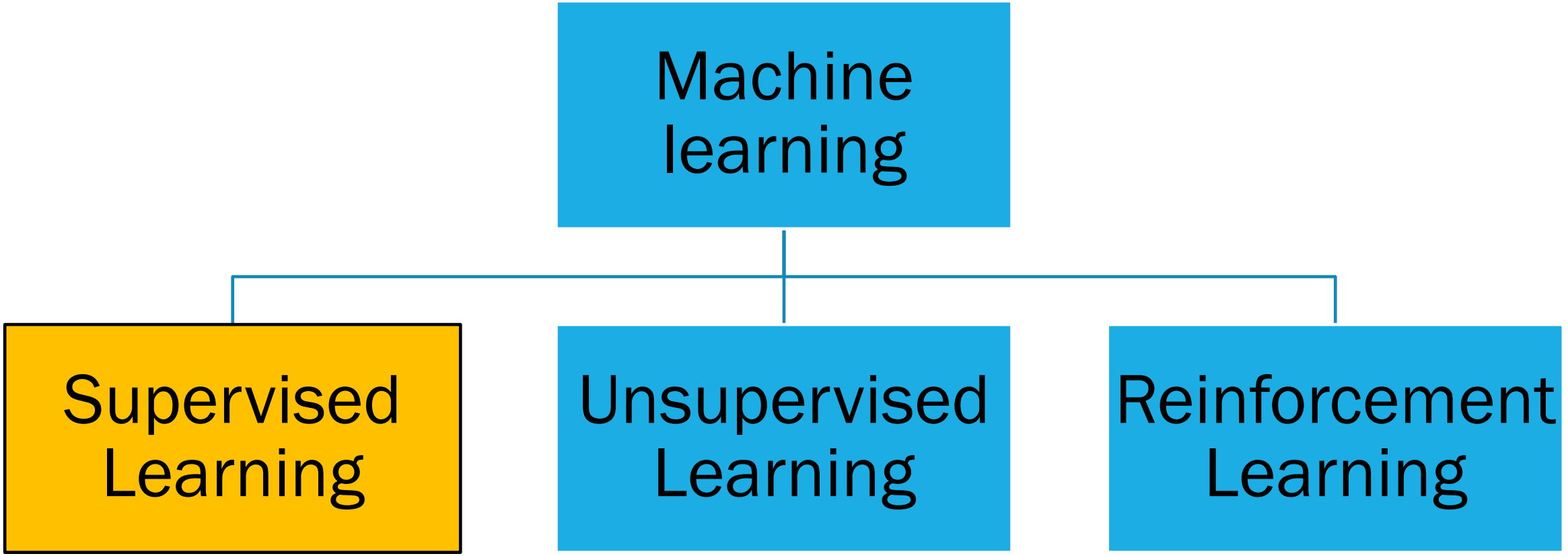
REVIEW Open Access

Deep learning in cancer diagnosis, prognosis and treatment selection

Khoa A. Tran[1,2], Olga Kondrashova[1], Andrew Bradley[4], Elizabeth D. Williams[2,3], John V. Pearson[1] and Nicola Waddell[1*]

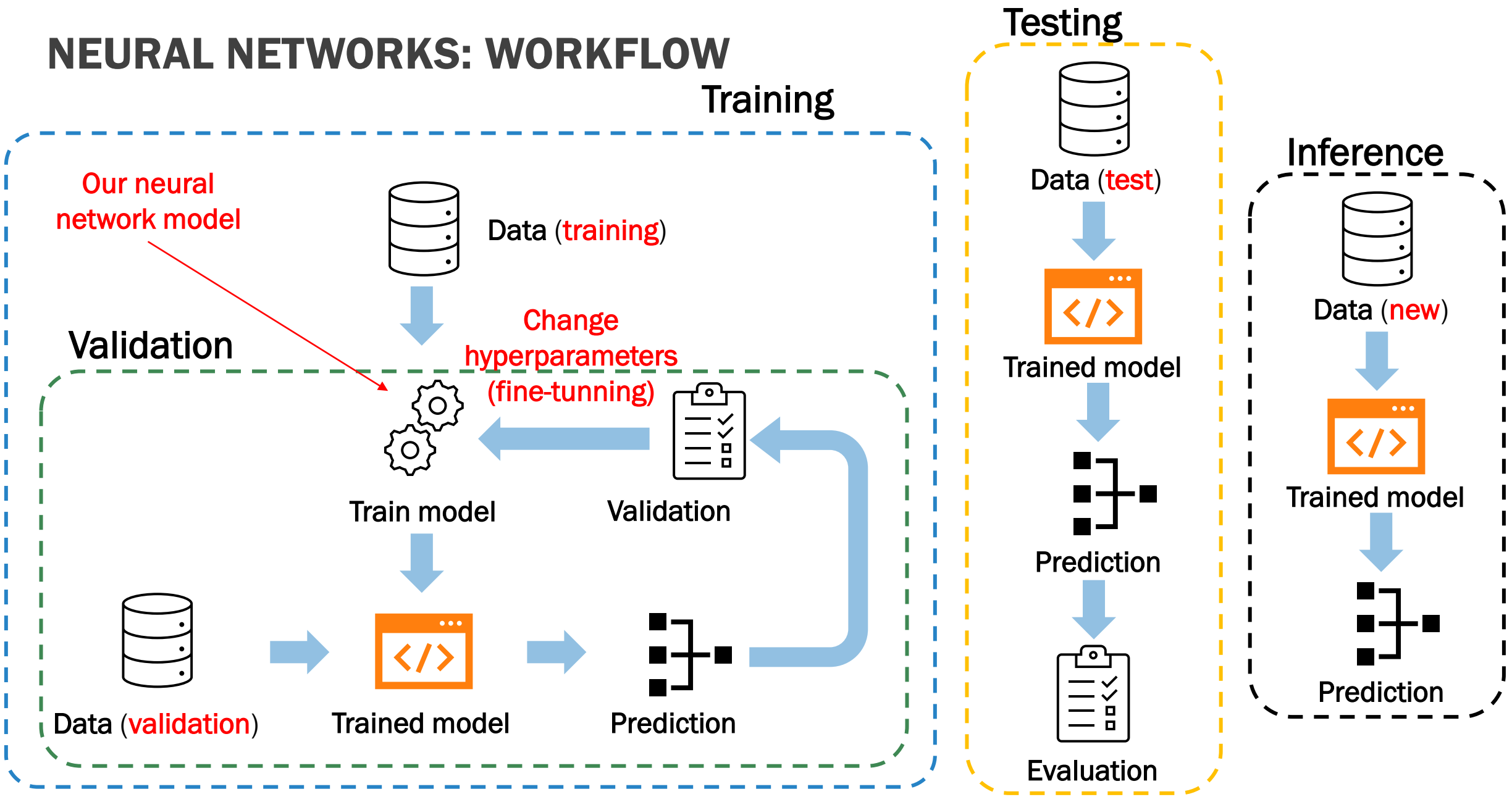
The content of this slide will become available after the lecture.

RECAP: TYPES OF MACHINE LEARNING



We have both **supervised** and **unsupervised** neural networks.
We will focus the rest of the lecture on **supervised** learning.

NEURAL NETWORKS: WORKFLOW



DATA SPLIT

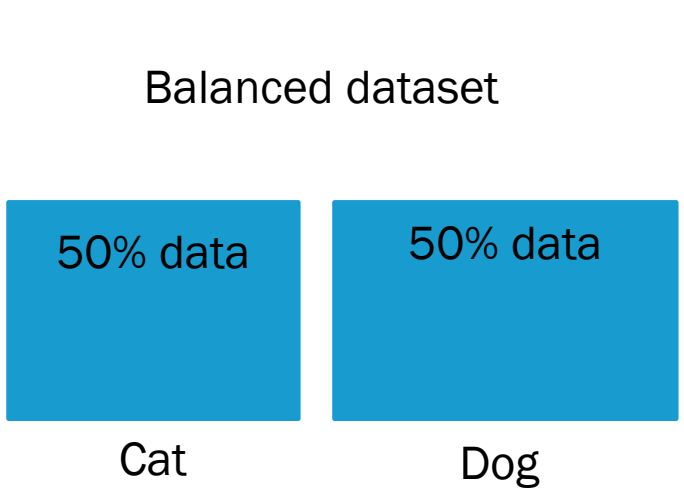
- Dataset:

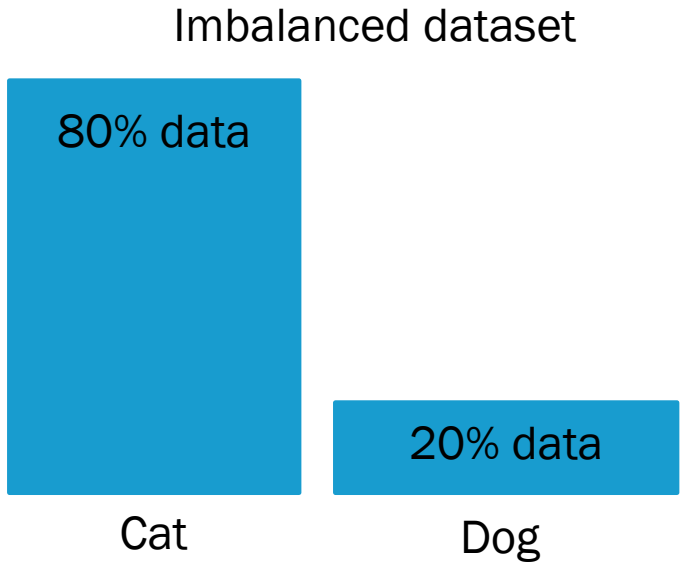


80% 10% 10% (small datasets)

60% 20% 20% (larger datasets)

- Know your data:

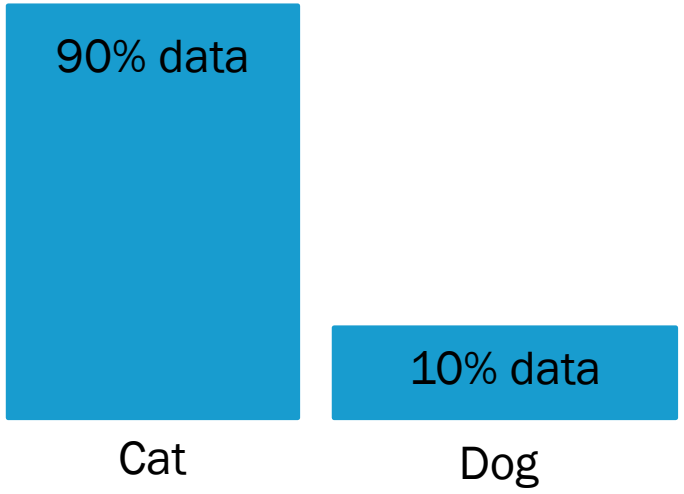




PROBLEMS WITH IMBALANCED DATASETS: ACCURACY

Prepare data (pre-processing)

Imbalanced dataset (100 samples)



Remember:
1) Know your data
2) Use and report accuracy for each class

What is for you a good accuracy?

If we predict everything as a cat….

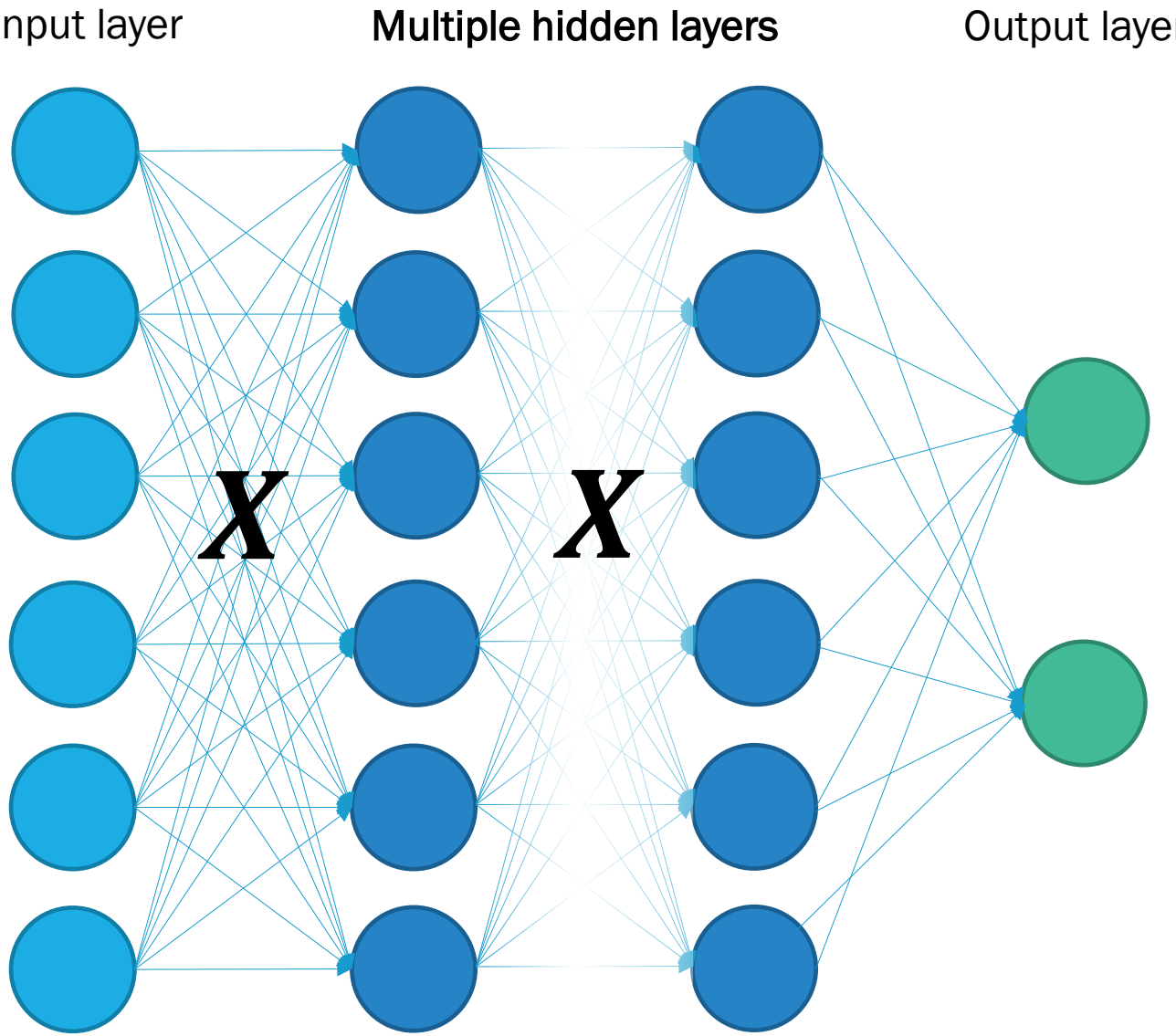
$$accuracy = \frac{number\ correct\ samples}{total\ number\ of\ samples}$$

$$accuracy = \frac{90}{100} = \mathbf{0.9}\ (\mathbf{90\%}!!)$$

This model would have a 90% accuracy even if predicts everything as a cat.

NEURAL NETWORK EXAMPLE

Input



How do we initialise weights?

How can we find the optimum weights so that the error is minimal?

TRAINING A NEURAL NETWORK

Train model

- Objective: find the optimum weights that will give the lowest **error** [**loss**]
- Optimisation (Gradient Descent)
  1. Randomly initialise weights
  2. Find how much the weights need to change (by computing the derivates)
  3. Update weights by taking a small step [**learning rate**]
  4. Repeat until convergence (until there is no improvement) [**number of epochs**]

FOR REFERENCE ONLY: LOSS OPTIMISATION

Train model

- **Identify (train) a set of weights that will give us the minimum error**
- Loss functions: Cross entropy, Mean Squared Error, Mean Absolute Error.
- Needs to be differentiable (the derivative of the function exists for all points).

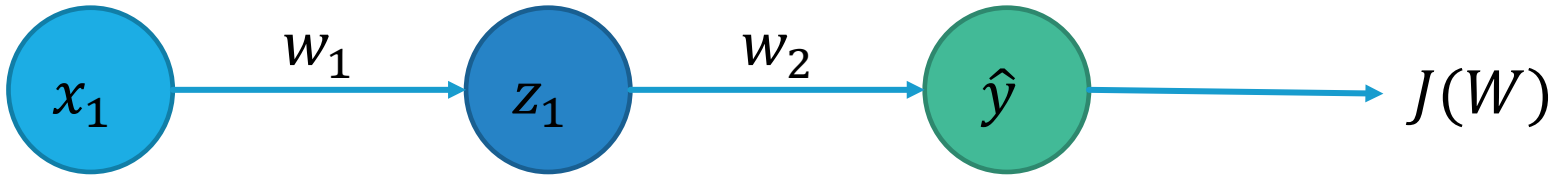
Binary classification	Multiclass classification	Regression
Loss: Cross Entropy	Loss: Cross Entropy	Loss: Mean Squared Error
Activation: Sigmoid	Activation: Softmax	

BACKPROPAGATION: BACKWARD PROPAGATION OF ERRORS

Train model

- Compute the gradients (computationally expensive!)

How does a small change on the weights affect the final loss $J(W)$?



Watch: https://www.youtube.com/watch?v=IN2XmBhILt4&list=PLblh5JKOoLUIxGDQs4LFFD--41Vzf-ME1&index=4

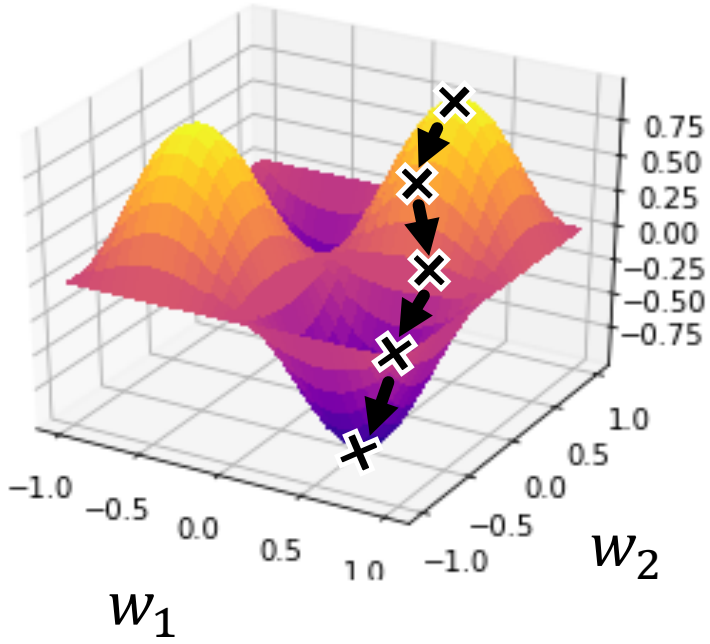
How much a change in w_2 will affect the total error (loss) $J(W)$?

The partial derivative of the loss with respect to w_2

$$\frac{\partial J(W)}{\partial w_2} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_2}$$

How much a change in w_1 will affect the total error (loss) $J(W)$?

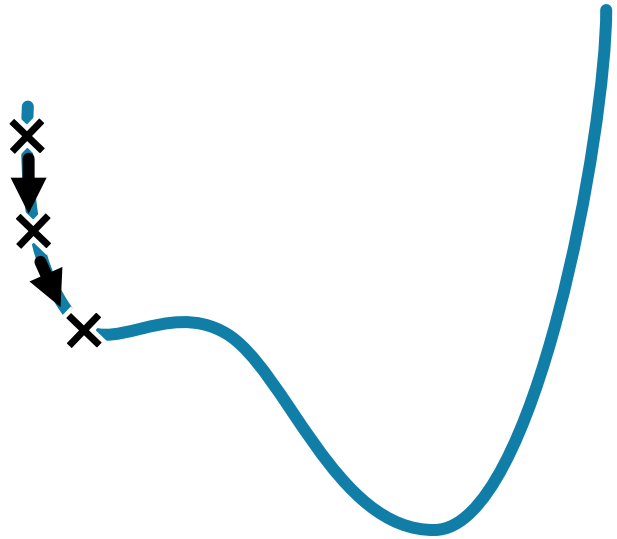
$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial z_1} * \frac{\partial z_1}{\partial w_1}$$



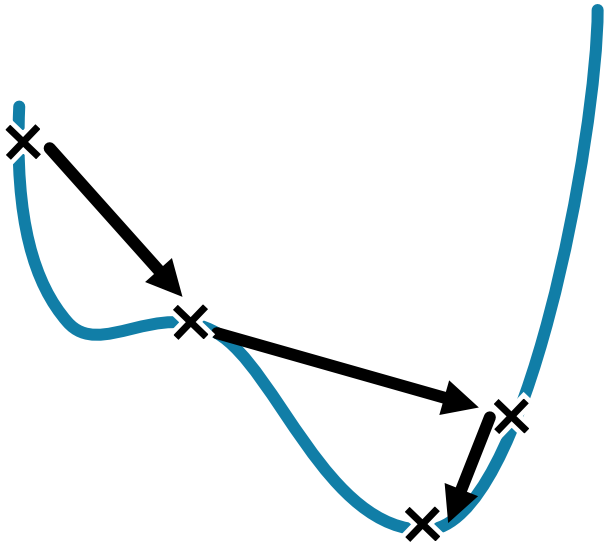
LEARNING RATE

Train model

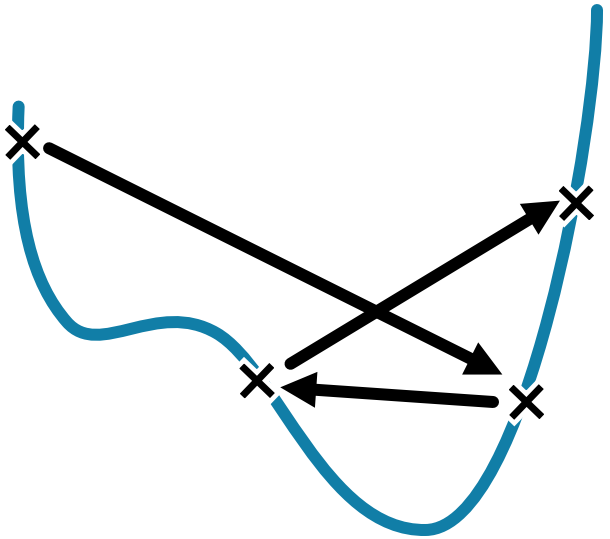
- How much step we take in the gradient?



Small learning rate



Good learning rate

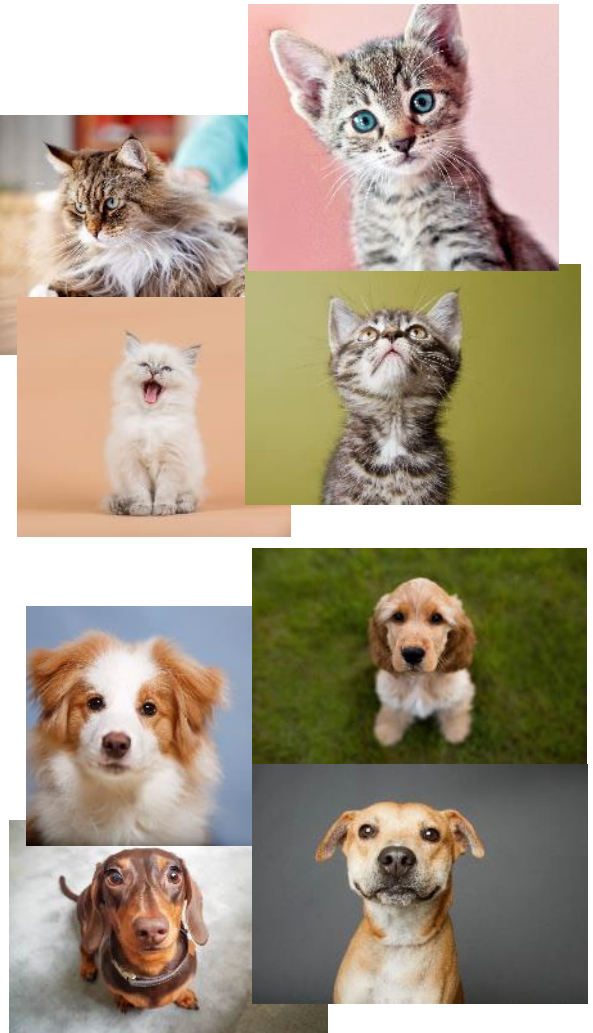


Large learning rate

FORWARD PROPAGATION AND BACKPROPAGATION

Train model

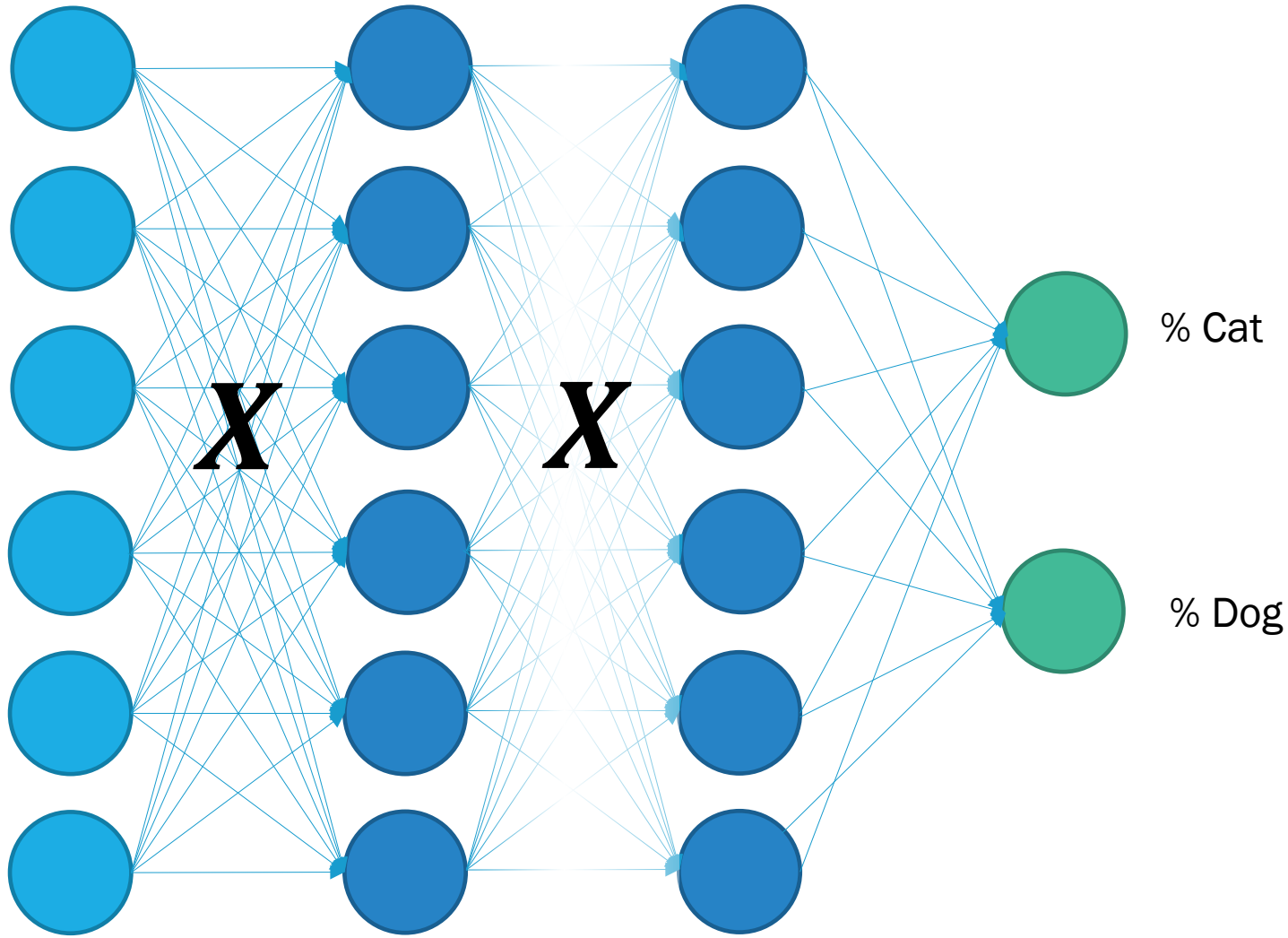
Input
(flatten)



Input layer

Multiple hidden layers

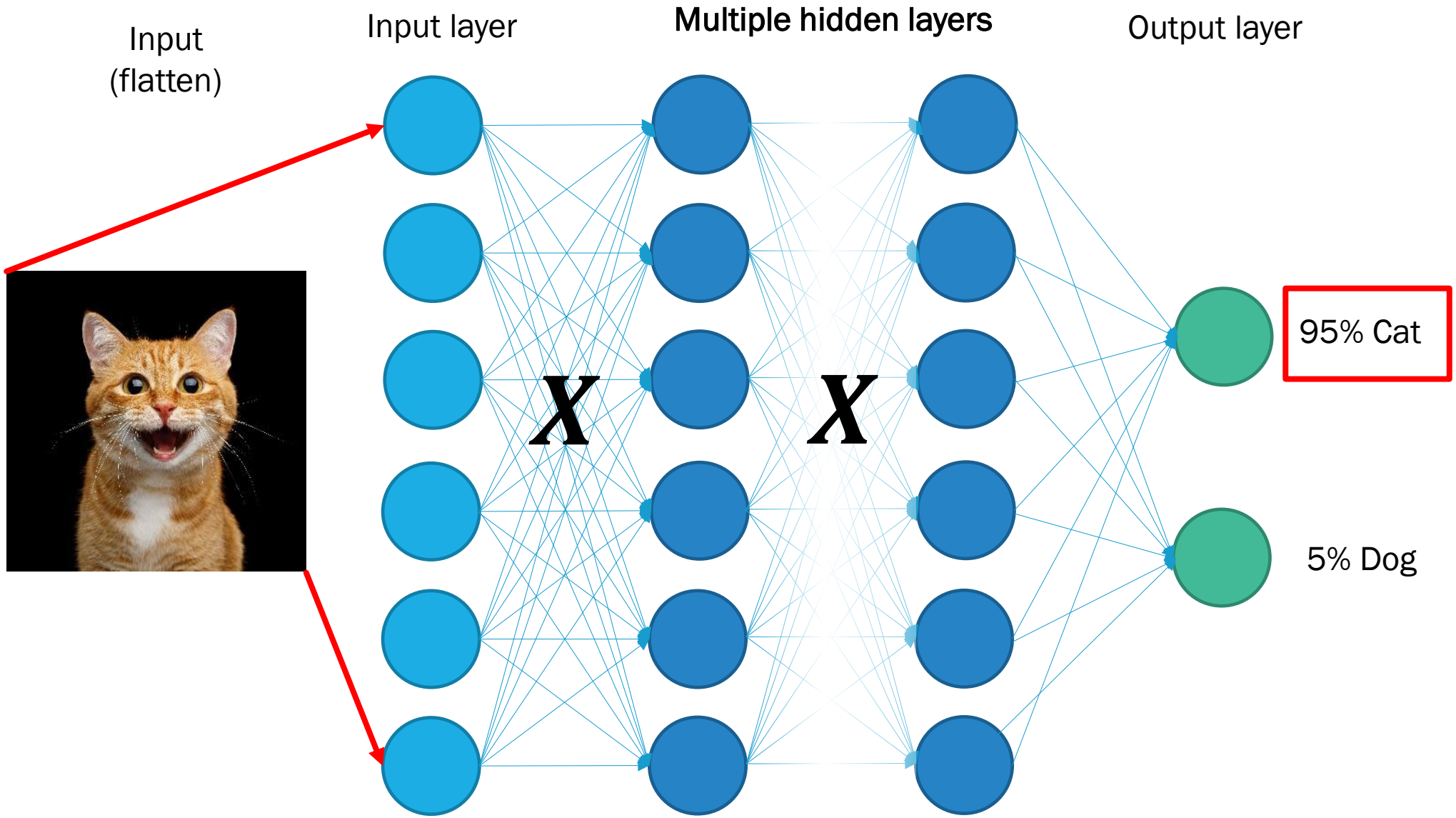
Output layer



% Cat

% Dog

FORWARD PROPAGATION AND BACKPROPAGATION



FORWARD PROPAGATION AND BACKPROPAGATION

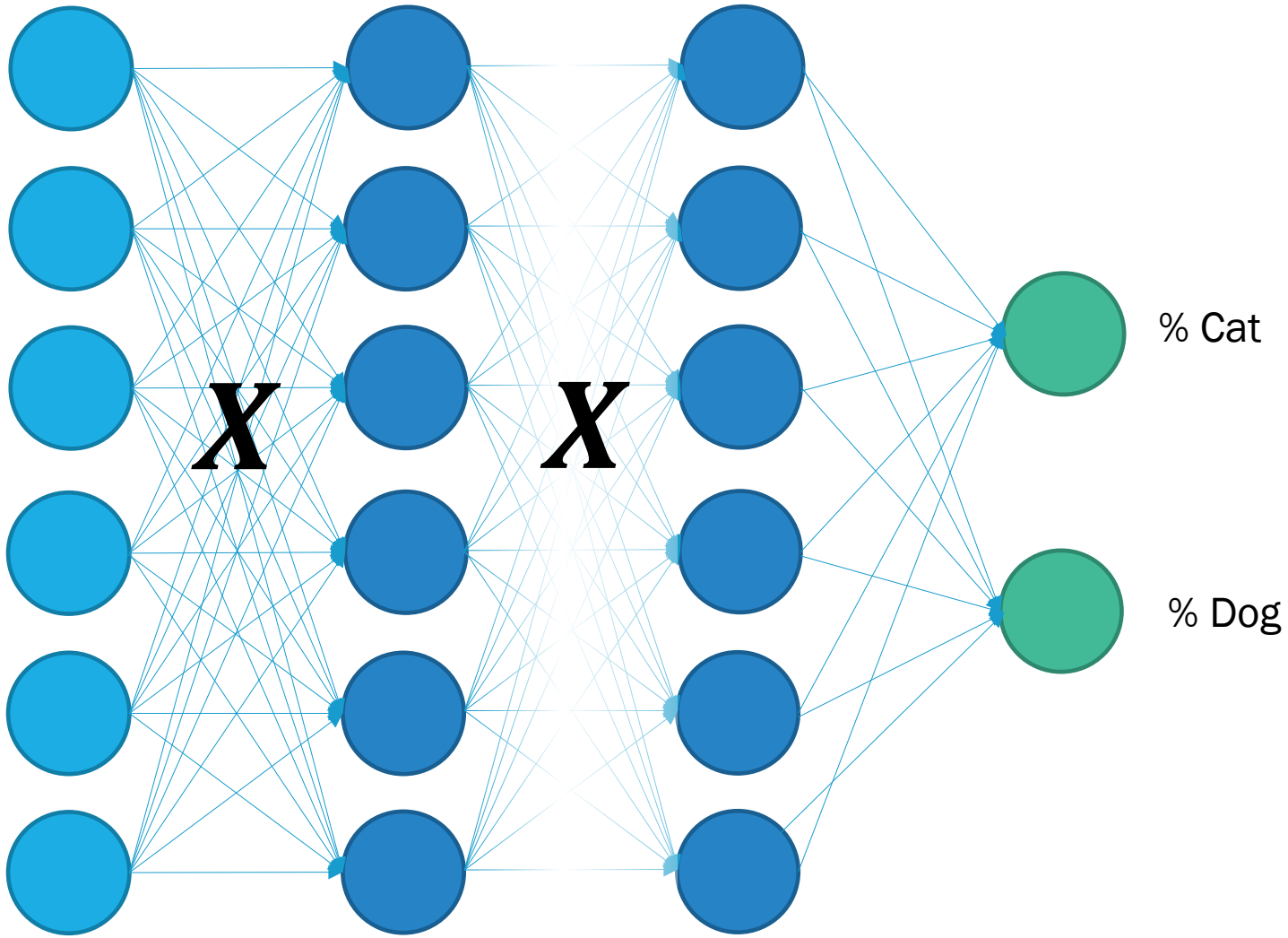
Input
(flatten)



Input layer

Multiple hidden layers

Output layer

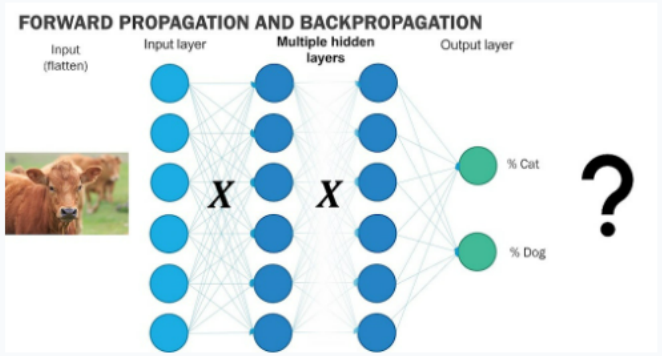


?

When poll is active respond at **PollEv.com/esterbonmati**

pollev.com/esterbonmati

What will the NN predict?



Will predict cow

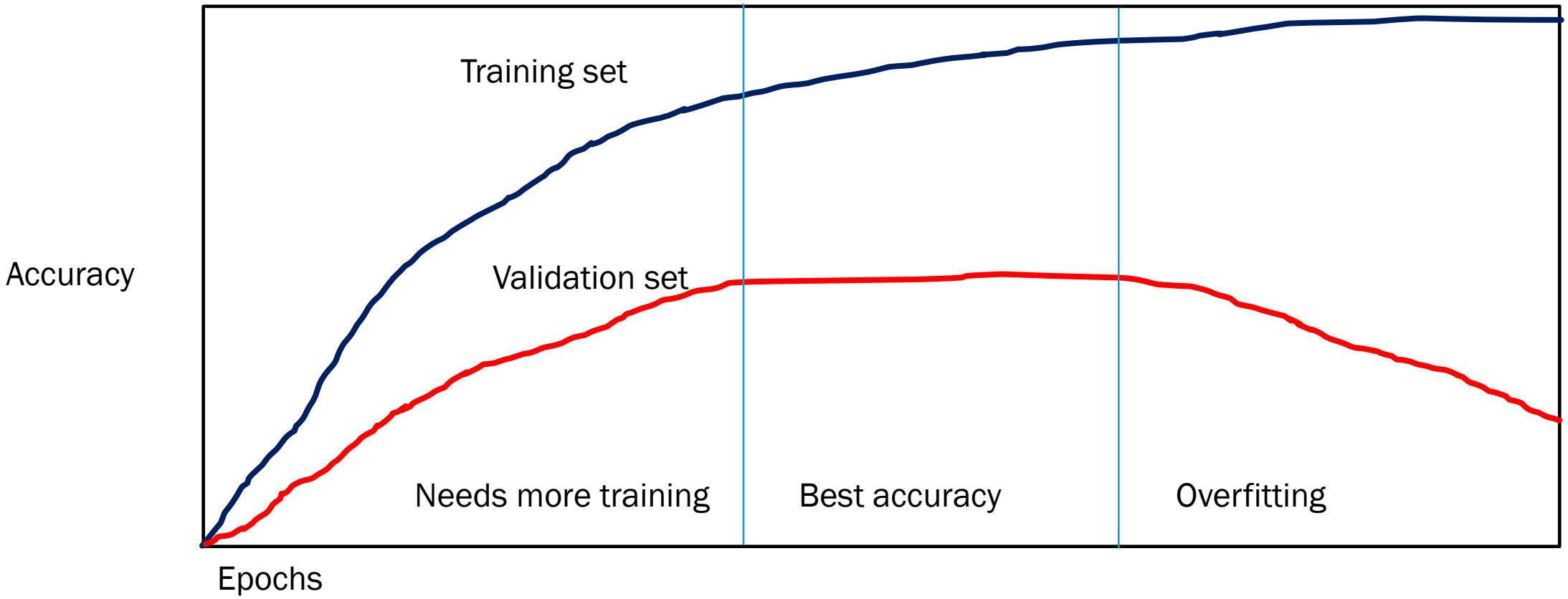
0%

SEE MORE

Powered by Poll Everywhere

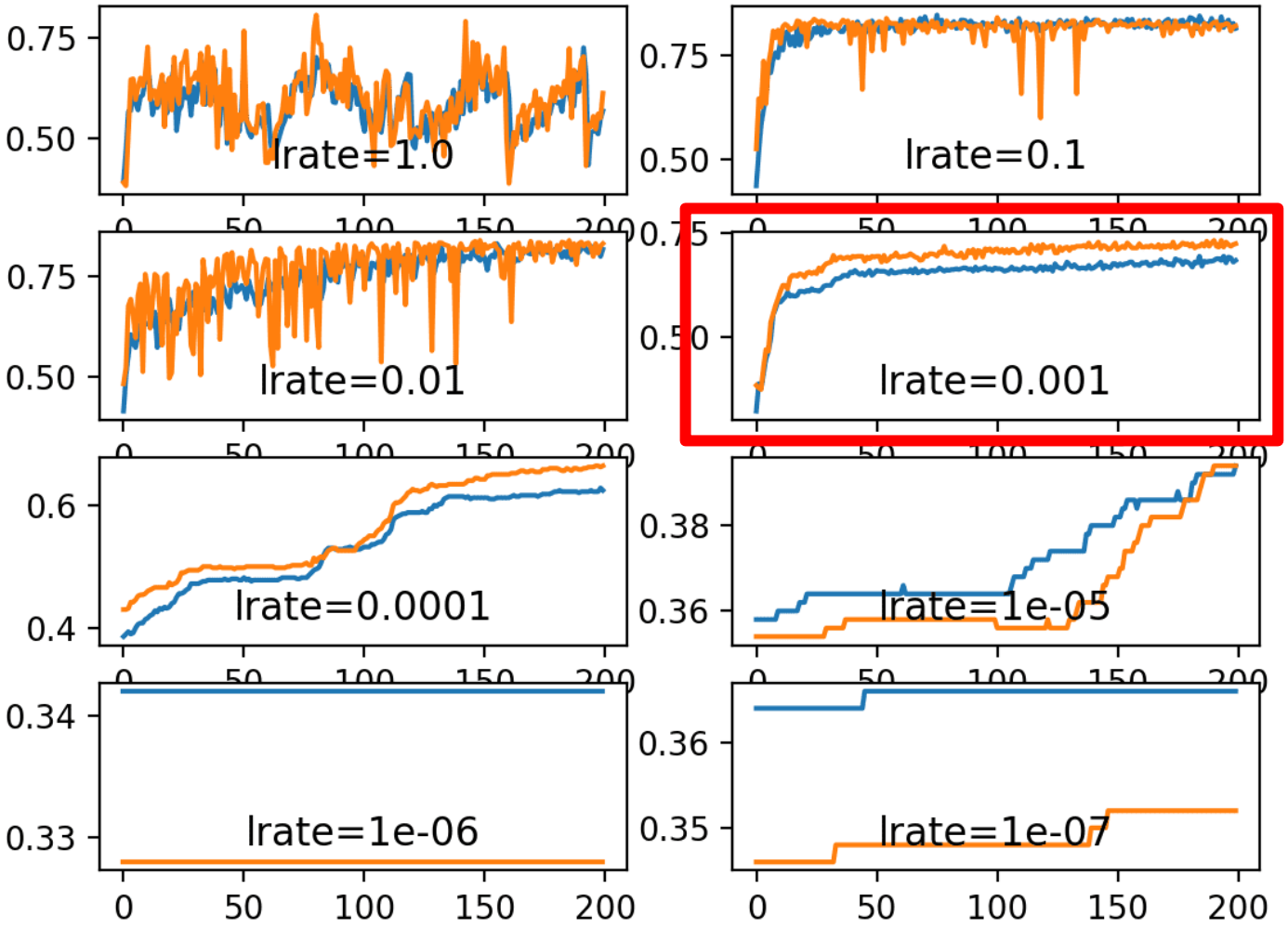
TRAINING

Train model



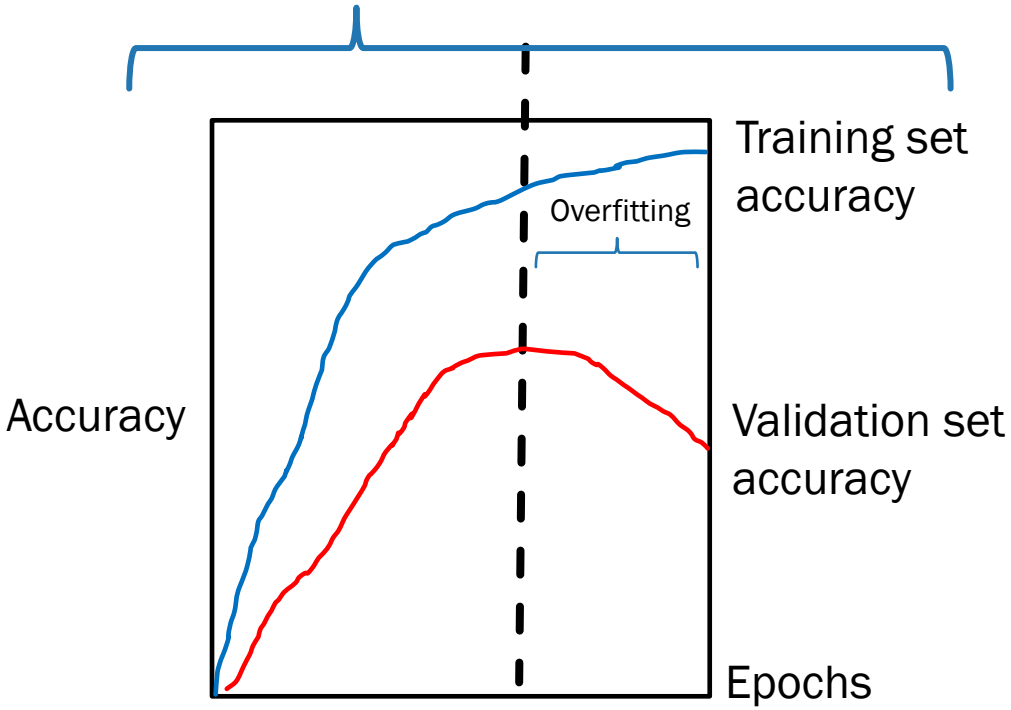
PROBLEMS: LEARNING RATE

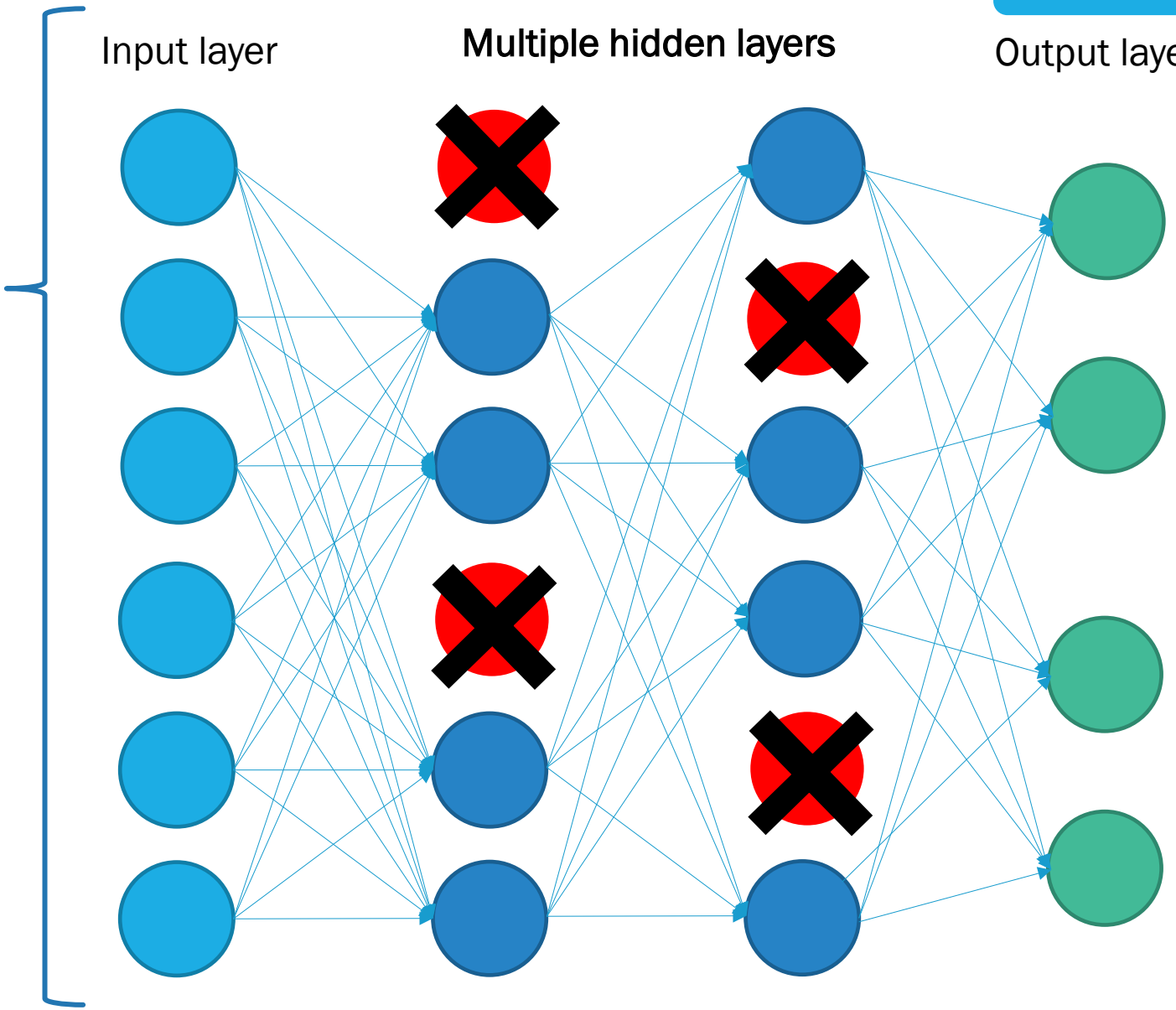
- If the learning rate is too high, the model may converge quickly and find a local minima
- If the learning rate is too small, the training can get stuck



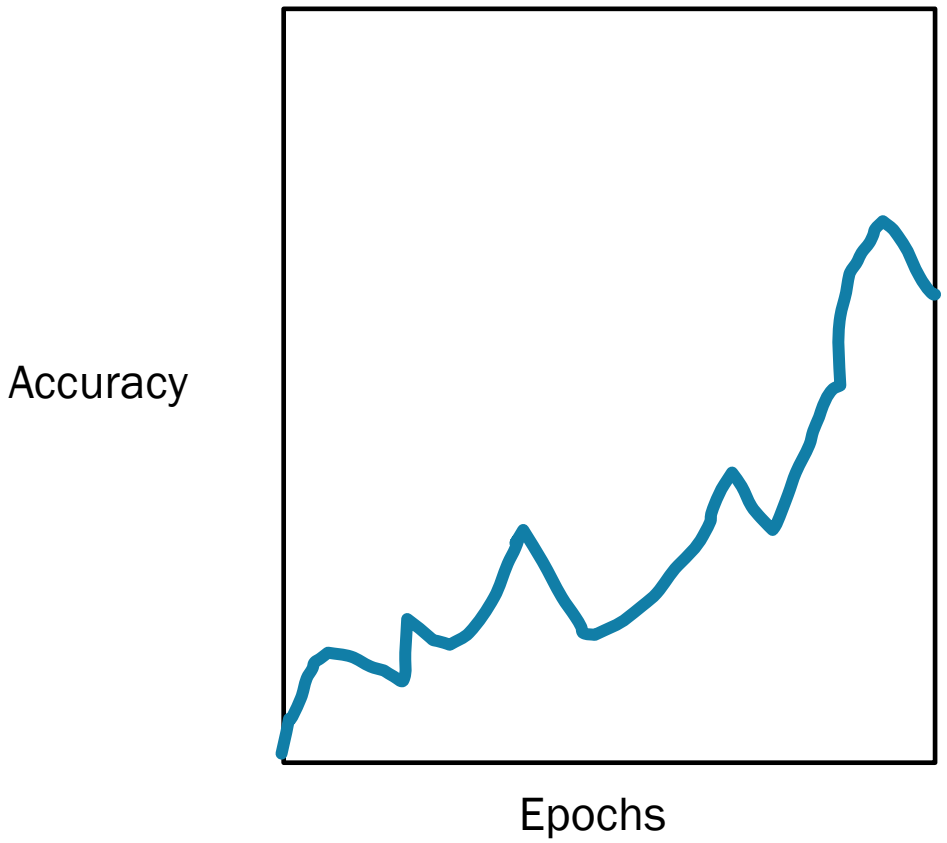
PROBLEMS: OVERFITTING

- When the network has 'memorised' the training data. Regularisation techniques:
  - **Dropout:** We set some activation to 0 (usually 50%) during training
  - **Early stopping**: stop training before it overfits the data





TRAINING



Applied AI

When poll is active respond at

PollEv.com /esterbonmati

What would you do next?

Decrease the learning rate

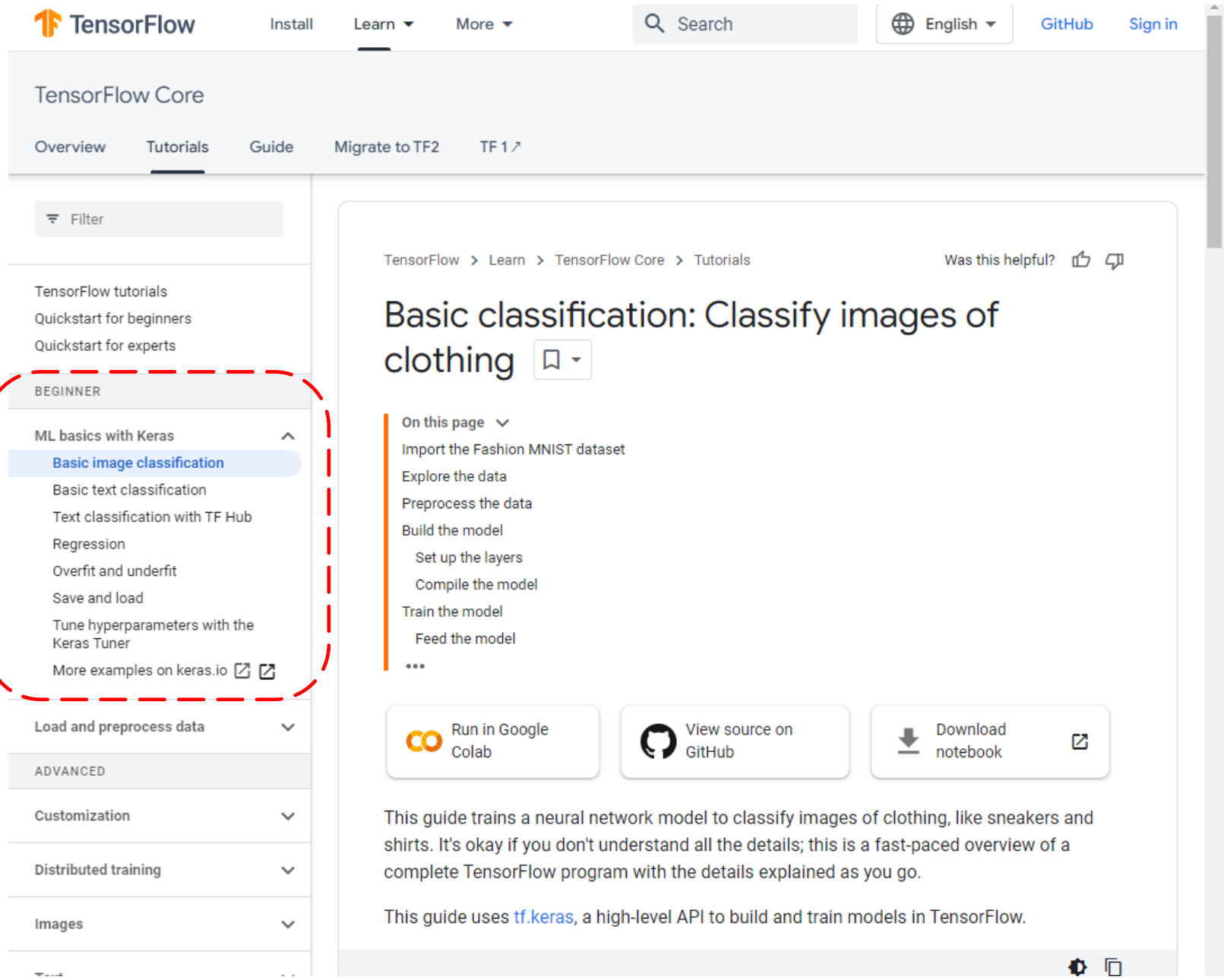
0%

Increase the learning rate

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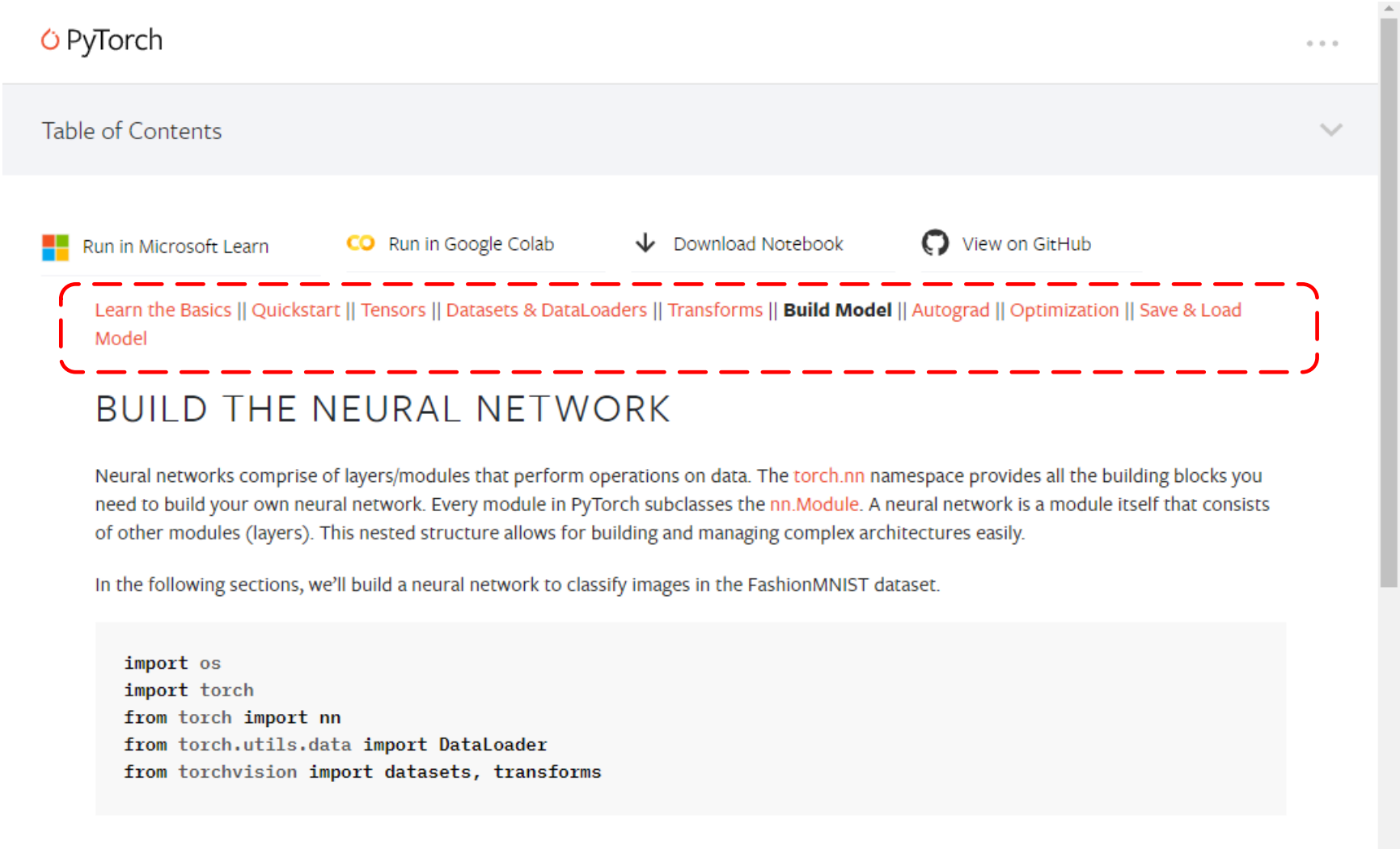
TENSORFLOW & KERAS

- https://www.tensorflow.org/tutorials/keras/classification



PYTORCH

- https://pytorch.org/tutorials/beginner/basics/buildmodel_tutorial.html



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Join by Web **PollEv.com/esterbonmati**

After: what is a neural network or artificial neural network?

Nobody has responded yet.

Hang tight! Responses are coming in.

Powered by Poll Everywhere

THANK YOU!